

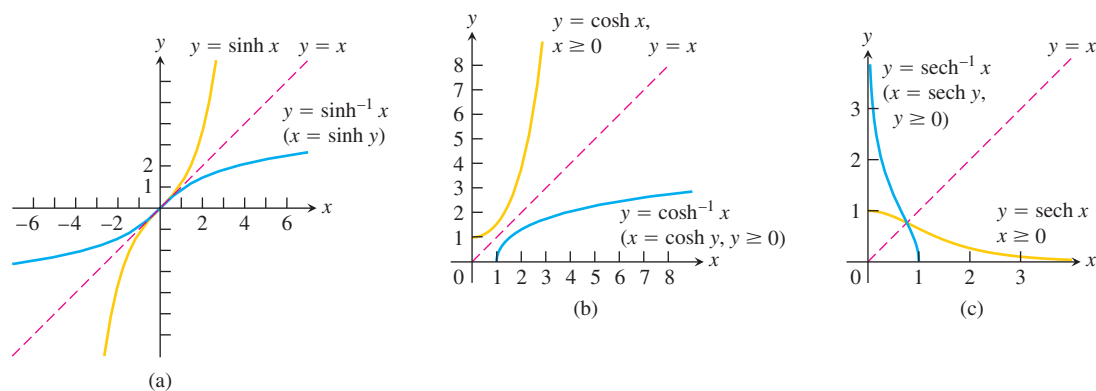


FIGURE 7.32 The graphs of the inverse hyperbolic sine, cosine, and secant of x . Notice the symmetries about the line $y = x$.

The hyperbolic tangent, cotangent, and cosecant are one-to-one on their domains and therefore have inverses, denoted by

$$y = \tanh^{-1} x, \quad y = \coth^{-1} x, \quad y = \operatorname{csch}^{-1} x.$$

These functions are graphed in Figure 7.33.

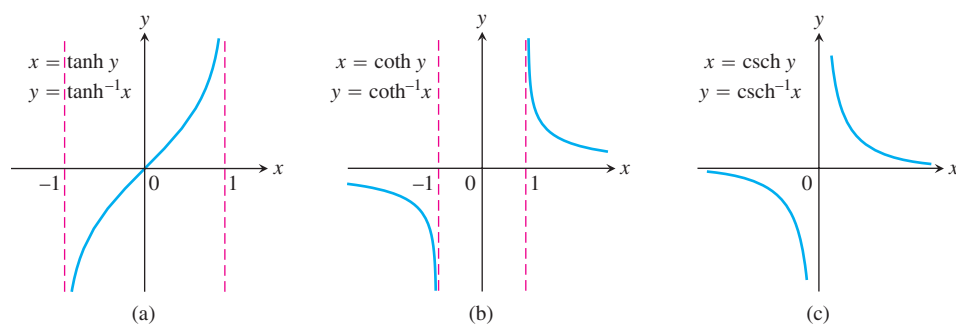


FIGURE 7.33 The graphs of the inverse hyperbolic tangent, cotangent, and cosecant of x .

TABLE 7.9 Identities for inverse hyperbolic functions

$$\operatorname{sech}^{-1} x = \cosh^{-1} \frac{1}{x}$$

$$\operatorname{csch}^{-1} x = \sinh^{-1} \frac{1}{x}$$

$$\coth^{-1} x = \tanh^{-1} \frac{1}{x}$$

Useful Identities

We use the identities in Table 7.9 to calculate the values of $\operatorname{sech}^{-1} x$, $\operatorname{csch}^{-1} x$, and $\coth^{-1} x$ on calculators that give only $\cosh^{-1} x$, $\sinh^{-1} x$, and $\tanh^{-1} x$. These identities are direct consequences of the definitions. For example, if $0 < x \leq 1$, then

$$\operatorname{sech} \left(\cosh^{-1} \left(\frac{1}{x} \right) \right) = \frac{1}{\cosh \left(\cosh^{-1} \left(\frac{1}{x} \right) \right)} = \frac{1}{\left(\frac{1}{x} \right)} = x$$

so

$$\cosh^{-1} \left(\frac{1}{x} \right) = \operatorname{sech}^{-1} x$$

since the hyperbolic secant is one-to-one on $(0, 1]$.

Derivatives and Integrals

The chief use of inverse hyperbolic functions lies in integrations that reverse the derivative formulas in Table 7.10.

TABLE 7.10 Derivatives of inverse hyperbolic functions

$$\begin{aligned} \frac{d(\sinh^{-1} u)}{dx} &= \frac{1}{\sqrt{1+u^2}} \frac{du}{dx} \\ \frac{d(\cosh^{-1} u)}{dx} &= \frac{1}{\sqrt{u^2-1}} \frac{du}{dx}, & u > 1 \\ \frac{d(\tanh^{-1} u)}{dx} &= \frac{1}{1-u^2} \frac{du}{dx}, & |u| < 1 \\ \frac{d(\coth^{-1} u)}{dx} &= \frac{1}{1-u^2} \frac{du}{dx}, & |u| > 1 \\ \frac{d(\operatorname{sech}^{-1} u)}{dx} &= \frac{-du/dx}{u\sqrt{1-u^2}}, & 0 < u < 1 \\ \frac{d(\operatorname{csch}^{-1} u)}{dx} &= \frac{-du/dx}{|u|\sqrt{1+u^2}}, & u \neq 0 \end{aligned}$$

The restrictions $|u| < 1$ and $|u| > 1$ on the derivative formulas for $\tanh^{-1} u$ and $\coth^{-1} u$ come from the natural restrictions on the values of these functions. (See Figure 7.33a and b.) The distinction between $|u| < 1$ and $|u| > 1$ becomes important when we convert the derivative formulas into integral formulas. If $|u| < 1$, the integral of $1/(1-u^2)$ is $\tanh^{-1} u + C$. If $|u| > 1$, the integral is $\coth^{-1} u + C$.

We illustrate how the derivatives of the inverse hyperbolic functions are found in Example 2, where we calculate $d(\cosh^{-1} u)/dx$. The other derivatives are obtained by similar calculations.

EXAMPLE 2 Derivative of the Inverse Hyperbolic Cosine

Show that if u is a differentiable function of x whose values are greater than 1, then

$$\frac{d}{dx} (\cosh^{-1} u) = \frac{1}{\sqrt{u^2-1}} \frac{du}{dx}.$$

HISTORICAL BIOGRAPHY

Sonya Kovalevsky
(1850–1891)

Solution First we find the derivative of $y = \cosh^{-1} x$ for $x > 1$ by applying Theorem 1 with $f(x) = \cosh x$ and $f^{-1}(x) = \cosh^{-1} x$. Theorem 1 can be applied because the derivative of $\cosh x$ is positive for $0 < x$.

$$\begin{aligned}
 (f^{-1})'(x) &= \frac{1}{f'(f^{-1}(x))} && \text{Theorem 1} \\
 &= \frac{1}{\sinh(\cosh^{-1} x)} && f'(u) = \sinh u \\
 &= \frac{1}{\sqrt{\cosh^2(\cosh^{-1} x) - 1}} && \cosh^2 u - \sinh^2 u = 1, \\
 & && \sinh u = \sqrt{\cosh^2 u - 1} \\
 &= \frac{1}{\sqrt{x^2 - 1}} && \cosh(\cosh^{-1} x) = x
 \end{aligned}$$

In short,

$$\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2 - 1}}.$$

The Chain Rule gives the final result:

$$\frac{d}{dx}(\cosh^{-1} u) = \frac{1}{\sqrt{u^2 - 1}} \frac{du}{dx}.$$

Instead of applying Theorem 1 directly, as in Example 2, we could also find the derivative of $y = \cosh^{-1} x$, $x > 1$, using implicit differentiation and the Chain Rule:

$$\begin{aligned}
 y &= \cosh^{-1} x \\
 x &= \cosh y && \text{Equivalent equation} \\
 1 &= \sinh y \frac{dy}{dx} && \text{Implicit differentiation} \\
 &&& \text{with respect to } x, \text{ and} \\
 &&& \text{the Chain Rule} \\
 \frac{dy}{dx} &= \frac{1}{\sinh y} = \frac{1}{\sqrt{\cosh^2 y - 1}} && \text{Since } x > 1, y > 0 \\
 &&& \text{and } \sinh y > 0 \\
 &= \frac{1}{\sqrt{x^2 - 1}}. && \cosh y = x
 \end{aligned}$$

With appropriate substitutions, the derivative formulas in Table 7.10 lead to the integration formulas in Table 7.11. Each of the formulas in Table 7.11 can be verified by differentiating the expression on the right-hand side.

EXAMPLE 3 Using Table 7.11

Evaluate

$$\int_0^1 \frac{2 \, dx}{\sqrt{3 + 4x^2}}.$$

TABLE 7.11 Integrals leading to inverse hyperbolic functions

1. $\int \frac{du}{\sqrt{a^2 + u^2}} = \sinh^{-1} \left(\frac{u}{a} \right) + C, \quad a > 0$
2. $\int \frac{du}{\sqrt{u^2 - a^2}} = \cosh^{-1} \left(\frac{u}{a} \right) + C, \quad u > a > 0$
3. $\int \frac{du}{a^2 - u^2} = \begin{cases} \frac{1}{a} \tanh^{-1} \left(\frac{u}{a} \right) + C & \text{if } u^2 < a^2 \\ \frac{1}{a} \coth^{-1} \left(\frac{u}{a} \right) + C, & \text{if } u^2 > a^2 \end{cases}$
4. $\int \frac{du}{u\sqrt{a^2 - u^2}} = -\frac{1}{a} \operatorname{sech}^{-1} \left(\frac{u}{a} \right) + C, \quad 0 < u < a$
5. $\int \frac{du}{u\sqrt{a^2 + u^2}} = -\frac{1}{a} \operatorname{csch}^{-1} \left| \frac{u}{a} \right| + C, \quad u \neq 0 \text{ and } a > 0$

Solution The indefinite integral is

$$\begin{aligned}
 \int \frac{2 \, dx}{\sqrt{3 + 4x^2}} &= \int \frac{du}{\sqrt{a^2 + u^2}} && u = 2x, \quad du = 2 \, dx, \quad a = \sqrt{3} \\
 &= \sinh^{-1} \left(\frac{u}{a} \right) + C && \text{Formula from Table 7.11} \\
 &= \sinh^{-1} \left(\frac{2x}{\sqrt{3}} \right) + C.
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \int_0^1 \frac{2 \, dx}{\sqrt{3 + 4x^2}} &= \sinh^{-1} \left(\frac{2x}{\sqrt{3}} \right) \Big|_0^1 = \sinh^{-1} \left(\frac{2}{\sqrt{3}} \right) - \sinh^{-1}(0) \\
 &= \sinh^{-1} \left(\frac{2}{\sqrt{3}} \right) - 0 \approx 0.98665.
 \end{aligned}$$

EXERCISES 7.8

Hyperbolic Function Values and Identities

Each of Exercises 1–4 gives a value of $\sinh x$ or $\cosh x$. Use the definitions and the identity $\cosh^2 x - \sinh^2 x = 1$ to find the values of the remaining five hyperbolic functions.

1. $\sinh x = -\frac{3}{4}$

2. $\sinh x = \frac{4}{3}$

3. $\cosh x = \frac{17}{15}, \quad x > 0$

4. $\cosh x = \frac{13}{5}, \quad x > 0$

Rewrite the expressions in Exercises 5–10 in terms of exponentials and simplify the results as much as you can.

5. $2 \cosh (\ln x)$

6. $\sinh (2 \ln x)$

7. $\cosh 5x + \sinh 5x$

8. $\cosh 3x - \sinh 3x$

9. $(\sinh x + \cosh x)^4$

10. $\ln (\cosh x + \sinh x) + \ln (\cosh x - \sinh x)$

11. Use the identities

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$$

$$\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$$

to show that

a. $\sinh 2x = 2 \sinh x \cosh x$

b. $\cosh 2x = \cosh^2 x + \sinh^2 x$.

12. Use the definitions of $\cosh x$ and $\sinh x$ to show that

$$\cosh^2 x - \sinh^2 x = 1.$$

Derivatives

In Exercises 13–24, find the derivative of y with respect to the appropriate variable.

13. $y = 6 \sinh \frac{x}{3}$

14. $y = \frac{1}{2} \sinh(2x + 1)$

15. $y = 2\sqrt{t} \tanh \sqrt{t}$

16. $y = t^2 \tanh \frac{1}{t}$

17. $y = \ln(\sinh z)$

18. $y = \ln(\cosh z)$

19. $y = \operatorname{sech} \theta(1 - \ln \operatorname{sech} \theta)$

20. $y = \operatorname{csch} \theta(1 - \ln \operatorname{csch} \theta)$

21. $y = \ln \cosh v - \frac{1}{2} \tanh^2 v$

22. $y = \ln \sinh v - \frac{1}{2} \coth^2 v$

23. $y = (x^2 + 1) \operatorname{sech}(\ln x)$

(Hint: Before differentiating, express in terms of exponentials and simplify.)

24. $y = (4x^2 - 1) \operatorname{csch}(\ln 2x)$

In Exercises 25–36, find the derivative of y with respect to the appropriate variable.

25. $y = \sinh^{-1} \sqrt{x}$

26. $y = \cosh^{-1} 2\sqrt{x+1}$

27. $y = (1 - \theta) \tanh^{-1} \theta$

28. $y = (\theta^2 + 2\theta) \tanh^{-1}(\theta + 1)$

29. $y = (1 - t) \coth^{-1} \sqrt{t}$

30. $y = (1 - t^2) \coth^{-1} t$

31. $y = \cos^{-1} x - x \operatorname{sech}^{-1} x$

32. $y = \ln x + \sqrt{1 - x^2} \operatorname{sech}^{-1} x$

33. $y = \operatorname{csch}^{-1} \left(\frac{1}{2} \right)^\theta$

34. $y = \operatorname{csch}^{-1} 2^\theta$

35. $y = \sinh^{-1}(\tan x)$

36. $y = \cosh^{-1}(\sec x), \quad 0 < x < \pi/2$

Integration Formulas

Verify the integration formulas in Exercises 37–40.

37. a. $\int \operatorname{sech} x \, dx = \tan^{-1}(\sinh x) + C$

b. $\int \operatorname{sech} x \, dx = \sin^{-1}(\tanh x) + C$

38. $\int x \operatorname{sech}^{-1} x \, dx = \frac{x^2}{2} \operatorname{sech}^{-1} x - \frac{1}{2} \sqrt{1 - x^2} + C$

39. $\int x \coth^{-1} x \, dx = \frac{x^2 - 1}{2} \coth^{-1} x + \frac{x}{2} + C$

40. $\int \tanh^{-1} x \, dx = x \tanh^{-1} x + \frac{1}{2} \ln(1 - x^2) + C$

Indefinite Integrals

Evaluate the integrals in Exercises 41–50.

41. $\int \sinh 2x \, dx$

42. $\int \sinh \frac{x}{5} \, dx$

43. $\int 6 \cosh \left(\frac{x}{2} - \ln 3 \right) dx$

44. $\int 4 \cosh(3x - \ln 2) \, dx$

45. $\int \tanh \frac{x}{7} \, dx$

46. $\int \coth \frac{\theta}{\sqrt{3}} \, d\theta$

47. $\int \operatorname{sech}^2 \left(x - \frac{1}{2} \right) dx$

48. $\int \operatorname{csch}^2(5 - x) \, dx$

49. $\int \frac{\operatorname{sech} \sqrt{t} \tanh \sqrt{t} \, dt}{\sqrt{t}}$

50. $\int \frac{\operatorname{csch}(\ln t) \coth(\ln t) \, dt}{t}$

Definite Integrals

Evaluate the integrals in Exercises 51–60.

51. $\int_{\ln 2}^{\ln 4} \coth x \, dx$

52. $\int_0^{\ln 2} \tanh 2x \, dx$

53. $\int_{-\ln 4}^{-\ln 2} 2e^\theta \cosh \theta \, d\theta$

54. $\int_0^{\ln 2} 4e^{-\theta} \sinh \theta \, d\theta$

55. $\int_{-\pi/4}^{\pi/4} \cosh(\tan \theta) \sec^2 \theta \, d\theta$

56. $\int_0^{\pi/2} 2 \sinh(\sin \theta) \cos \theta \, d\theta$

57. $\int_1^2 \frac{\cosh(\ln t)}{t} \, dt$

58. $\int_1^4 \frac{8 \cosh \sqrt{x}}{\sqrt{x}} \, dx$

59. $\int_{-\ln 2}^0 \cosh^2 \left(\frac{x}{2} \right) dx$

60. $\int_0^{\ln 10} 4 \sinh^2 \left(\frac{x}{2} \right) dx$

Evaluating Inverse Hyperbolic Functions and Related Integrals

When hyperbolic function keys are not available on a calculator, it is still possible to evaluate the inverse hyperbolic functions by expressing them as logarithms, as shown here.

$$\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1}), \quad -\infty < x < \infty$$

$$\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1}), \quad x \geq 1$$

$$\tanh^{-1} x = \frac{1}{2} \ln \frac{1+x}{1-x}, \quad |x| < 1$$

$$\operatorname{sech}^{-1} x = \ln \left(\frac{1 + \sqrt{1 - x^2}}{x} \right), \quad 0 < x \leq 1$$

$$\operatorname{csch}^{-1} x = \ln \left(\frac{1}{x} + \frac{\sqrt{1 + x^2}}{|x|} \right), \quad x \neq 0$$

$$\coth^{-1} x = \frac{1}{2} \ln \frac{x+1}{x-1}, \quad |x| > 1$$

Use the formulas in the box here to express the numbers in Exercises 61–66 in terms of natural logarithms.

61. $\sinh^{-1}(-5/12)$ 62. $\cosh^{-1}(5/3)$
 63. $\tanh^{-1}(-1/2)$ 64. $\coth^{-1}(5/4)$
 65. $\operatorname{sech}^{-1}(3/5)$ 66. $\operatorname{csch}^{-1}(-1/\sqrt{3})$

Evaluate the integrals in Exercises 67–74 in terms of

- a. inverse hyperbolic functions.
 b. natural logarithms.

67. $\int_0^{2\sqrt{3}} \frac{dx}{\sqrt{4+x^2}}$ 68. $\int_0^{1/3} \frac{6 dx}{\sqrt{1+9x^2}}$
 69. $\int_{5/4}^2 \frac{dx}{1-x^2}$ 70. $\int_0^{1/2} \frac{dx}{1-x^2}$
 71. $\int_{1/5}^{3/13} \frac{dx}{x\sqrt{1-16x^2}}$ 72. $\int_1^2 \frac{dx}{x\sqrt{4+x^2}}$
 73. $\int_0^\pi \frac{\cos x dx}{\sqrt{1+\sin^2 x}}$ 74. $\int_1^e \frac{dx}{x\sqrt{1+(\ln x)^2}}$

Applications and Theory

75. a. Show that if a function f is defined on an interval symmetric about the origin (so that f is defined at $-x$ whenever it is defined at x), then

$$f(x) = \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2}. \quad (1)$$

Then show that $(f(x) + f(-x))/2$ is even and that $(f(x) - f(-x))/2$ is odd.

- b. Equation (1) simplifies considerably if f itself is (i) even or (ii) odd. What are the new equations? Give reasons for your answers.
76. Derive the formula $\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$, $-\infty < x < \infty$. Explain in your derivation why the plus sign is used with the square root instead of the minus sign.
77. **Skydiving** If a body of mass m falling from rest under the action of gravity encounters an air resistance proportional to the square of the velocity, then the body's velocity t sec into the fall satisfies the differential equation

$$m \frac{dv}{dt} = mg - kv^2,$$

where k is a constant that depends on the body's aerodynamic properties and the density of the air. (We assume that the fall is short enough so that the variation in the air's density will not affect the outcome significantly.)

- a. Show that

$$v = \sqrt{\frac{mg}{k}} \tanh\left(\sqrt{\frac{gk}{m}} t\right)$$

satisfies the differential equation and the initial condition that $v = 0$ when $t = 0$.

- b. Find the body's *limiting velocity*, $\lim_{t \rightarrow \infty} v$.
 c. For a 160-lb skydiver ($mg = 160$), with time in seconds and distance in feet, a typical value for k is 0.005. What is the diver's limiting velocity?

78. **Accelerations whose magnitudes are proportional to displacement** Suppose that the position of a body moving along a coordinate line at time t is

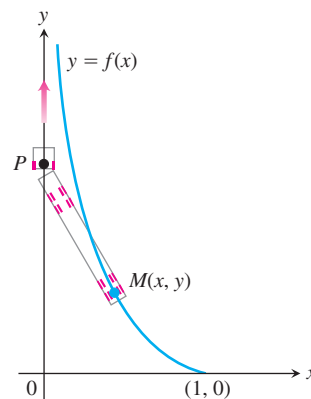
- a. $s = a \cos kt + b \sin kt$
 b. $s = a \cosh kt + b \sinh kt$.

Show in both cases that the acceleration d^2s/dt^2 is proportional to s but that in the first case it is directed toward the origin, whereas in the second case it is directed away from the origin.

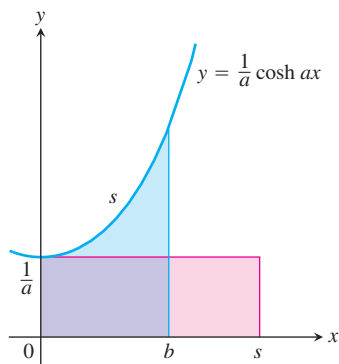
79. **Tractor trailers and the tractrix** When a tractor trailer turns into a cross street or driveway, its rear wheels follow a curve like the one shown here. (This is why the rear wheels sometimes ride up over the curb.) We can find an equation for the curve if we picture the rear wheels as a mass M at the point $(1, 0)$ on the x -axis attached by a rod of unit length to a point P representing the cab at the origin. As the point P moves up the y -axis, it drags M along behind it. The curve traced by M —called a *tractrix* from the Latin word *tractum*, for “drag”—can be shown to be the graph of the function $y = f(x)$ that solves the initial value problem

Differential equation: $\frac{dy}{dx} = -\frac{1}{x\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}}$
 Initial condition: $y = 0$ when $x = 1$.

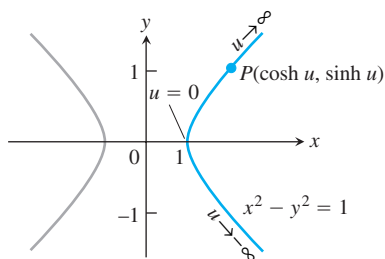
Solve the initial value problem to find an equation for the curve. (You need an inverse hyperbolic function.)



80. **Area** Show that the area of the region in the first quadrant enclosed by the curve $y = (1/a) \cosh ax$, the coordinate axes, and the line $x = b$ is the same as the area of a rectangle of height $1/a$ and length s , where s is the length of the curve from $x = 0$ to $x = b$. (See accompanying figure.)



- 81. Volume** A region in the first quadrant is bounded above by the curve $y = \cosh x$, below by the curve $y = \sinh x$, and on the left and right by the y -axis and the line $x = 2$, respectively. Find the volume of the solid generated by revolving the region about the x -axis.
- 82. Volume** The region enclosed by the curve $y = \operatorname{sech} x$, the x -axis, and the lines $x = \pm \ln \sqrt{3}$ is revolved about the x -axis to generate a solid. Find the volume of the solid.
- 83. Arc length** Find the length of the segment of the curve $y = (1/2) \cosh 2x$ from $x = 0$ to $x = \ln \sqrt{5}$.
- 84. The hyperbolic in hyperbolic functions** In case you are wondering where the name *hyperbolic* comes from, here is the answer: Just as $x = \cos u$ and $y = \sin u$ are identified with points (x, y) on the unit circle, the functions $x = \cosh u$ and $y = \sinh u$ are identified with points (x, y) on the right-hand branch of the unit hyperbola, $x^2 - y^2 = 1$.



Since $\cosh^2 u - \sinh^2 u = 1$, the point $(\cosh u, \sinh u)$ lies on the right-hand branch of the hyperbola $x^2 - y^2 = 1$ for every value of u (Exercise 84).

Another analogy between hyperbolic and circular functions is that the variable u in the coordinates $(\cosh u, \sinh u)$ for the points of the right-hand branch of the hyperbola $x^2 - y^2 = 1$ is twice the area of the sector AOP pictured in the accompanying figure. To see why this is so, carry out the following steps.

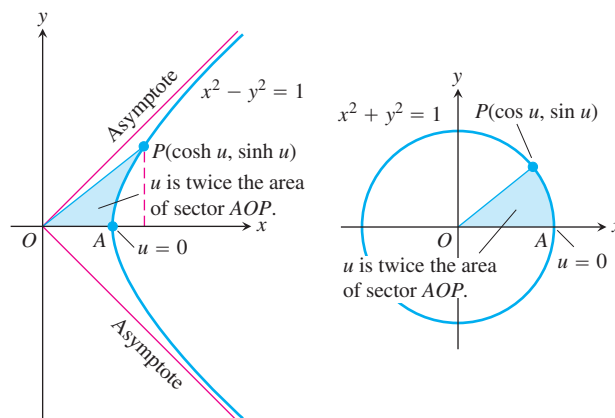
- a.** Show that the area $A(u)$ of sector AOP is

$$A(u) = \frac{1}{2} \cosh u \sinh u - \int_1^{\cosh u} \sqrt{x^2 - 1} \, dx.$$

- b.** Differentiate both sides of the equation in part (a) with respect to u to show that

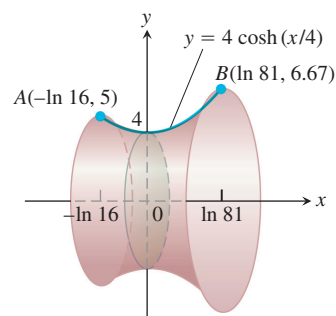
$$A'(u) = \frac{1}{2}.$$

- c.** Solve this last equation for $A(u)$. What is the value of $A(0)$? What is the value of the constant of integration C in your solution? With C determined, what does your solution say about the relationship of u to $A(u)$?



One of the analogies between hyperbolic and circular functions is revealed by these two diagrams (Exercise 84).

- 85. A minimal surface** Find the area of the surface swept out by revolving about the x -axis the curve $y = 4 \cosh(x/4)$, $-\ln 16 \leq x \leq \ln 81$.



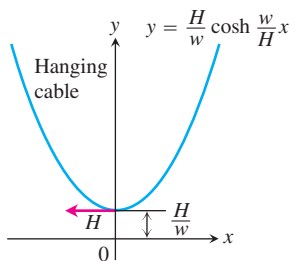
It can be shown that, of all continuously differentiable curves joining points A and B in the figure, the curve $y = 4 \cosh(x/4)$ generates the surface of least area. If you made a rigid wire frame of the end-circles through A and B and dipped them in a soap-film solution, the surface spanning the circles would be the one generated by the curve.

- T 86. a.** Find the centroid of the curve $y = \cosh x$, $-\ln 2 \leq x \leq \ln 2$.
- b.** Evaluate the coordinates to two decimal places. Then sketch the curve and plot the centroid to show its relation to the curve.

Hanging Cables

87. Imagine a cable, like a telephone line or TV cable, strung from one support to another and hanging freely. The cable's weight per unit length is w and the horizontal tension at its lowest point is a vector of length H . If we choose a coordinate system for the plane of the cable in which the x -axis is horizontal, the force of gravity is straight down, the positive y -axis points straight up, and the lowest point of the cable lies at the point $y = H/w$ on the y -axis (see accompanying figure), then it can be shown that the cable lies along the graph of the hyperbolic cosine

$$y = \frac{H}{w} \cosh \frac{w}{H} x.$$

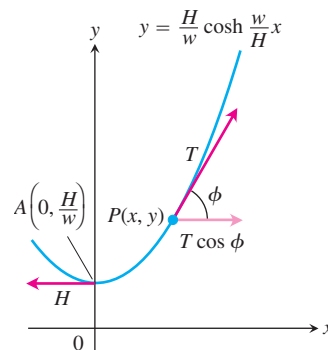


Such a curve is sometimes called a **chain curve** or a **catenary**, the latter deriving from the Latin *catena*, meaning “chain.”

- a. Let $P(x, y)$ denote an arbitrary point on the cable. The next accompanying figure displays the tension at P as a vector of length (magnitude) T , as well as the tension H at the lowest point A . Show that the cable's slope at P is

$$\tan \phi = \frac{dy}{dx} = \sinh \frac{w}{H} x.$$

- b. Using the result from part (a) and the fact that the horizontal tension at P must equal H (the cable is not moving), show that $T = wy$. Hence, the magnitude of the tension at $P(x, y)$ is exactly equal to the weight of y units of cable.



88. (Continuation of Exercise 87.) The length of arc AP in the Exercise 87 figure is $s = (1/a) \sinh ax$, where $a = w/H$. Show that the coordinates of P may be expressed in terms of s as

$$x = \frac{1}{a} \sinh^{-1} as, \quad y = \sqrt{s^2 + \frac{1}{a^2}}.$$

89. **The sag and horizontal tension in a cable** The ends of a cable 32 ft long and weighing 2 lb/ft are fastened at the same level to posts 30 ft apart.

- a. Model the cable with the equation

$$y = \frac{1}{a} \cosh ax, \quad -15 \leq x \leq 15.$$

Use information from Exercise 88 to show that a satisfies the equation

$$16a = \sinh 15a. \quad (2)$$

- T** b. Solve Equation (2) graphically by estimating the coordinates of the points where the graphs of the equations $y = 16a$ and $y = \sinh 15a$ intersect in the ay -plane.
- T** c. Solve Equation (2) for a numerically. Compare your solution with the value you found in part (b).
- d. Estimate the horizontal tension in the cable at the cable's lowest point.
- T** e. Using the value found for a in part (c), graph the catenary

$$y = \frac{1}{a} \cosh ax$$

over the interval $-15 \leq x \leq 15$. Estimate the sag in the cable at its center.

Chapter 7 Additional and Advanced Exercises

Limits

Find the limits in Exercises 1–6.

$$1. \lim_{b \rightarrow 1^-} \int_0^b \frac{dx}{\sqrt{1-x^2}}$$

$$2. \lim_{x \rightarrow \infty} \frac{1}{x} \int_0^x \tan^{-1} t \, dt$$

$$3. \lim_{x \rightarrow 0^+} (\cos \sqrt{x})^{1/x}$$

$$4. \lim_{x \rightarrow \infty} (x + e^x)^{2/x}$$

$$5. \lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} \right)$$

$$6. \lim_{n \rightarrow \infty} \frac{1}{n} (e^{1/n} + e^{2/n} + \cdots + e^{(n-1)/n} + e^{n/n})$$

7. Let $A(t)$ be the area of the region in the first quadrant enclosed by the coordinate axes, the curve $y = e^{-x}$, and the vertical line $x = t$, $t > 0$. Let $V(t)$ be the volume of the solid generated by revolving the region about the x -axis. Find the following limits.

$$\text{a. } \lim_{t \rightarrow \infty} A(t) \quad \text{b. } \lim_{t \rightarrow \infty} V(t)/A(t) \quad \text{c. } \lim_{t \rightarrow 0^+} V(t)/A(t)$$

8. Varying a logarithm's base

a. Find $\lim \log_a 2$ as $a \rightarrow 0^+$, 1^- , 1^+ , and ∞ .

T b. Graph $y = \log_a 2$ as a function of a over the interval $0 < a \leq 4$.

Theory and Examples

9. Find the areas between the curves $y = 2(\log_2 x)/x$ and $y = 2(\log_4 x)/x$ and the x -axis from $x = 1$ to $x = e$. What is the ratio of the larger area to the smaller?

T 10. Graph $f(x) = \tan^{-1} x + \tan^{-1}(1/x)$ for $-5 \leq x \leq 5$. Then use calculus to explain what you see. How would you expect f to behave beyond the interval $[-5, 5]$? Give reasons for your answer.

11. For what $x > 0$ does $x^{(x^x)} = (x^x)^x$? Give reasons for your answer.

T 12. Graph $f(x) = (\sin x)^{\sin x}$ over $[0, 3\pi]$. Explain what you see.

13. Find $f'(2)$ if $f(x) = e^{g(x)}$ and $g(x) = \int_2^x \frac{t}{1+t^4} dt$.

14. a. Find df/dx if

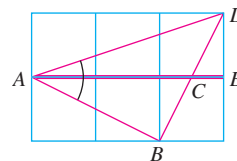
$$f(x) = \int_1^{e^x} \frac{2 \ln t}{t} dt.$$

b. Find $f(0)$.

c. What can you conclude about the graph of f ? Give reasons for your answer.

15. The figure here shows an informal proof that

$$\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \frac{\pi}{4}.$$

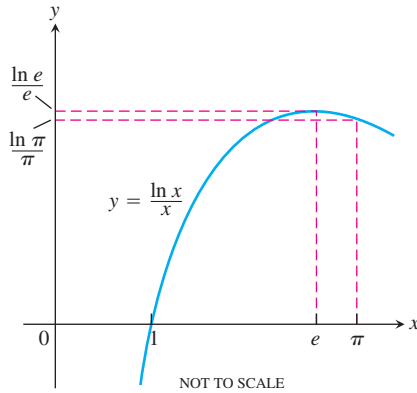


How does the argument go? (Source: “Behold! Sums of Arctan,” by Edward M. Harris, *College Mathematics Journal*, Vol. 18, No. 2, Mar. 1987, p. 141.)

16. $\pi^e < e^\pi$

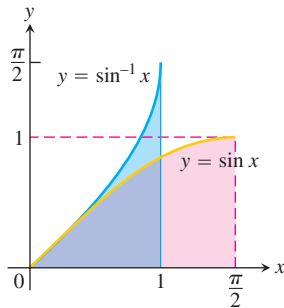
a. Why does the accompanying figure “prove” that $\pi^e < e^\pi$? (Source: “Proof Without Words,” by Fouad Nakhil, *Mathematics Magazine*, Vol. 60, No. 3, June 1987, p. 165.)

b. The accompanying figure assumes that $f(x) = (\ln x)/x$ has an absolute maximum value at $x = e$. How do you know it does?



17. Use the accompanying figure to show that

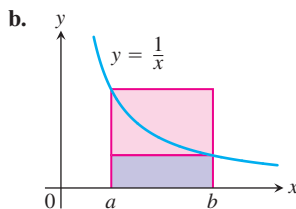
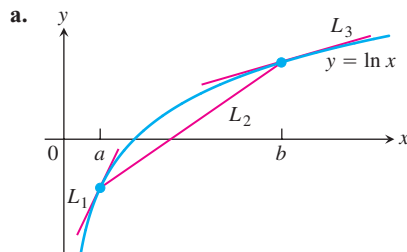
$$\int_0^{\pi/2} \sin x \, dx = \frac{\pi}{2} - \int_0^1 \sin^{-1} x \, dx.$$



18. **Napier's inequality** Here are two pictorial proofs that

$$b > a > 0 \Rightarrow \frac{1}{b} < \frac{\ln b - \ln a}{b - a} < \frac{1}{a}.$$

Explain what is going on in each case.



(Source: Roger B. Nelson, *College Mathematics Journal*, Vol. 24, No. 2, March 1993, p. 165.)

19. Even-odd decompositions

- Suppose that g is an even function of x and h is an odd function of x . Show that if $g(x) + h(x) = 0$ for all x then $g(x) = 0$ for all x and $h(x) = 0$ for all x .
- Use the result in part (a) to show that if $f(x) = f_E(x) + f_O(x)$ is the sum of an even function $f_E(x)$ and an odd function $f_O(x)$, then

$$f_E(x) = (f(x) + f(-x))/2 \quad \text{and} \quad f_O(x) = (f(x) - f(-x))/2.$$

- What is the significance of the result in part (b)?
20. Let g be a function that is differentiable throughout an open interval containing the origin. Suppose g has the following properties:

- $g(x + y) = \frac{g(x) + g(y)}{1 - g(x)g(y)}$ for all real numbers x, y , and $x + y$ in the domain of g .

- $\lim_{h \rightarrow 0} g(h) = 0$

- $\lim_{h \rightarrow 0} \frac{g(h)}{h} = 1$

- Show that $g(0) = 0$.
- Show that $g'(x) = 1 + [g(x)]^2$.
- Find $g(x)$ by solving the differential equation in part (b).

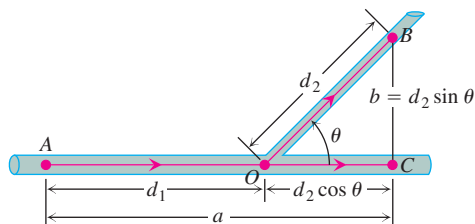
Applications

- Center of mass** Find the center of mass of a thin plate of constant density covering the region in the first and fourth quadrants enclosed by the curves $y = 1/(1 + x^2)$ and $y = -1/(1 + x^2)$ and by the lines $x = 0$ and $x = 1$.
- Solid of revolution** The region between the curve $y = 1/(2\sqrt{x})$ and the x -axis from $x = 1/4$ to $x = 4$ is revolved about the x -axis to generate a solid.
 - Find the volume of the solid.
 - Find the centroid of the region.
- The Rule of 70** If you use the approximation $\ln 2 \approx 0.70$ (in place of $0.69314\dots$), you can derive a rule of thumb that says, "To estimate how many years it will take an amount of money to double when invested at r percent compounded continuously, divide r into 70." For instance, an amount of money invested at 5% will double in about $70/5 = 14$ years. If you want it to double in 10 years instead, you have to invest it at $70/10 = 7\%$. Show how the Rule of 70 is derived. (A similar "Rule of 72" uses 72 instead of 70, because 72 has more integer factors.)
- Free fall in the fourteenth century** In the middle of the fourteenth century, Albert of Saxony (1316–1390) proposed a model of free fall that assumed that the velocity of a falling body was proportional to the distance fallen. It seemed reasonable to think that a body that had fallen 20 ft might be moving twice as fast as a body that had fallen 10 ft. And besides, none of the instruments in use at the time were accurate enough to prove otherwise. Today we can see just how far off Albert of Saxony's model was by

solving the initial value problem implicit in his model. Solve the problem and compare your solution graphically with the equation $s = 16t^2$. You will see that it describes a motion that starts too slowly at first and then becomes too fast too soon to be realistic.

- 25. The best branching angles for blood vessels and pipes** When a smaller pipe branches off from a larger one in a flow system, we may want it to run off at an angle that is best from some energy-saving point of view. We might require, for instance, that energy loss due to friction be minimized along the section AOB shown in the accompanying figure. In this diagram, B is a given point to be reached by the smaller pipe, A is a point in the larger pipe upstream from B , and O is the point where the branching occurs. A law due to Poiseuille states that the loss of energy due to friction in nonturbulent flow is proportional to the length of the path and inversely proportional to the fourth power of the radius. Thus, the loss along AO is $(kd_1)/R^4$ and along OB is $(kd_2)/r^4$, where k is a constant, d_1 is the length of AO , d_2 is the length of OB , R is the radius of the larger pipe, and r is the radius of the smaller pipe. The angle θ is to be chosen to minimize the sum of these two losses:

$$L = k \frac{d_1}{R^4} + k \frac{d_2}{r^4}.$$



In our model, we assume that $AC = a$ and $BC = b$ are fixed. Thus we have the relations

$$d_1 + d_2 \cos \theta = a \quad d_2 \sin \theta = b,$$

so that

$$\begin{aligned} d_2 &= b \csc \theta, \\ d_1 &= a - d_2 \cos \theta = a - b \cot \theta. \end{aligned}$$

We can express the total loss L as a function of θ :

$$L = k \left(\frac{a - b \cot \theta}{R^4} + \frac{b \csc \theta}{r^4} \right).$$

- a. Show that the critical value of θ for which $dL/d\theta$ equals zero is

$$\theta_c = \cos^{-1} \frac{r^4}{R^4}.$$

- b. If the ratio of the pipe radii is $r/R = 5/6$, estimate to the nearest degree the optimal branching angle given in part (a).

The mathematical analysis described here is also used to explain the angles at which arteries branch in an animal's body. (See *Introduction to Mathematics for Life Scientists*, Second Edition, by E. Batschelet [New York: Springer-Verlag, 1976].)

- T 26. Group blood testing** During World War II it was necessary to administer blood tests to large numbers of recruits. There are two standard ways to administer a blood test to N people. In method 1, each person is tested separately. In method 2, the blood samples of x people are pooled and tested as one large sample. If the test is negative, this one test is enough for all x people. If the test is positive, then each of the x people is tested separately, requiring a total of $x + 1$ tests. Using the second method and some probability theory it can be shown that, on the average, the total number of tests y will be

$$y = N \left(1 - q^x + \frac{1}{x} \right).$$

With $q = 0.99$ and $N = 1000$, find the integer value of x that minimizes y . Also find the integer value of x that maximizes y . (This second result is not important to the real-life situation.) The group testing method was used in World War II with a savings of 80% over the individual testing method, but not with the given value of q .

Chapter 7

Practice Exercises

Differentiation

In Exercises 1–24, find the derivative of y with respect to the appropriate variable.

1. $y = 10e^{-x/5}$
2. $y = \sqrt{2}e^{\sqrt{2}x}$
3. $y = \frac{1}{4}xe^{4x} - \frac{1}{16}e^{4x}$
4. $y = x^2e^{-2/x}$
5. $y = \ln(\sin^2 \theta)$
6. $y = \ln(\sec^2 \theta)$
7. $y = \log_2(x^2/2)$
8. $y = \log_5(3x - 7)$
9. $y = 8^{-t}$
10. $y = 9^{2t}$
11. $y = 5x^{3.6}$
12. $y = \sqrt{2}x^{-\sqrt{2}}$
13. $y = (x + 2)^{x+2}$
14. $y = 2(\ln x)^{x/2}$
15. $y = \sin^{-1}\sqrt{1 - u^2}, \quad 0 < u < 1$
16. $y = \sin^{-1}\left(\frac{1}{\sqrt{v}}\right), \quad v > 1$
17. $y = \ln \cos^{-1} x$
18. $y = z \cos^{-1} z - \sqrt{1 - z^2}$
19. $y = t \tan^{-1} t - \frac{1}{2} \ln t$
20. $y = (1 + t^2) \cot^{-1} 2t$

$$21. y = z \sec^{-1} z - \sqrt{z^2 - 1}, \quad z > 1$$

$$22. y = 2\sqrt{x-1} \sec^{-1}\sqrt{x}$$

$$23. y = \csc^{-1}(\sec \theta), \quad 0 < \theta < \pi/2$$

$$24. y = (1 + x^2)e^{\tan^{-1} x}$$

Logarithmic Differentiation

In Exercises 25–30, use logarithmic differentiation to find the derivative of y with respect to the appropriate variable.

$$25. y = \frac{2(x^2 + 1)}{\sqrt{\cos 2x}} \qquad 26. y = \frac{10\sqrt{3x+4}}{\sqrt{2x-4}}$$

$$27. y = \left(\frac{(t+1)(t-1)}{(t-2)(t+3)} \right)^5, \quad t > 2$$

$$28. y = \frac{2u2^u}{\sqrt{u^2 + 1}}$$

$$29. y = (\sin \theta)^{\sqrt{\theta}} \qquad 30. y = (\ln x)^{1/(\ln x)}$$

Integration

Evaluate the integrals in Exercises 31–78.

$$31. \int e^x \sin(e^x) dx$$

$$32. \int e^t \cos(3e^t - 2) dt$$

33. $\int e^x \sec^2(e^x - 7) dx$

34. $\int e^y \csc(e^y + 1) \cot(e^y + 1) dy$

35. $\int \sec^2(x) e^{\tan x} dx$

37. $\int_{-1}^1 \frac{dx}{3x - 4}$

39. $\int_0^\pi \tan \frac{x}{3} dx$

41. $\int_0^4 \frac{2t}{t^2 - 25} dt$

43. $\int \frac{\tan(\ln v)}{v} dv$

45. $\int \frac{(\ln x)^{-3}}{x} dx$

47. $\int \frac{1}{r} \csc^2(1 + \ln r) dr$

49. $\int x 3^{x^2} dx$

51. $\int_1^7 \frac{3}{x} dx$

53. $\int_1^4 \left(\frac{x}{8} + \frac{1}{2x} \right) dx$

55. $\int_{-2}^{-1} e^{-(x+1)} dx$

57. $\int_0^{\ln 5} e^r (3e^r + 1)^{-3/2} dr$

59. $\int_1^e \frac{1}{x} (1 + 7 \ln x)^{-1/3} dx$

61. $\int_1^3 \frac{(\ln(v+1))^2}{v+1} dv$

63. $\int_1^8 \frac{\log_4 \theta}{\theta} d\theta$

65. $\int_{-3/4}^{3/4} \frac{6 dx}{\sqrt{9 - 4x^2}}$

67. $\int_{-2}^2 \frac{3 dt}{4 + 3t^2}$

69. $\int \frac{dy}{y\sqrt{4y^2 - 1}}$

71. $\int_{\sqrt{2}/3}^{2/3} \frac{dy}{|y|\sqrt{9y^2 - 1}}$

73. $\int \frac{dx}{\sqrt{-2x - x^2}}$

36. $\int \csc^2 x e^{\cot x} dx$

38. $\int_1^e \frac{\sqrt{\ln x}}{x} dx$

40. $\int_{1/6}^{1/4} 2 \cot \pi x dx$

42. $\int_{-\pi/2}^{\pi/6} \frac{\cos t}{1 - \sin t} dt$

44. $\int \frac{dv}{v \ln v}$

46. $\int \frac{\ln(x-5)}{x-5} dx$

48. $\int \frac{\cos(1 - \ln v)}{v} dv$

50. $\int 2^{\tan x} \sec^2 x dx$

52. $\int_1^{32} \frac{1}{5x} dx$

54. $\int_1^8 \left(\frac{2}{3x} - \frac{8}{x^2} \right) dx$

56. $\int_{-\ln 2}^0 e^{2w} dw$

58. $\int_0^{\ln 9} e^\theta (e^\theta - 1)^{1/2} d\theta$

60. $\int_e^{e^2} \frac{1}{x\sqrt{\ln x}} dx$

62. $\int_2^4 (1 + \ln t) t \ln t dt$

64. $\int_1^e \frac{8 \ln 3 \log_3 \theta}{\theta} d\theta$

66. $\int_{-1/5}^{1/5} \frac{6 dx}{\sqrt{4 - 25x^2}}$

68. $\int_{\sqrt{3}}^3 \frac{dt}{3 + t^2}$

70. $\int \frac{24 dy}{y\sqrt{y^2 - 16}}$

72. $\int_{-2/\sqrt{5}}^{-\sqrt{6}/\sqrt{5}} \frac{dy}{|y|\sqrt{5y^2 - 3}}$

74. $\int \frac{dx}{\sqrt{-x^2 + 4x - 1}}$

75. $\int_{-2}^{-1} \frac{2 dv}{v^2 + 4v + 5}$

77. $\int \frac{dt}{(t+1)\sqrt{t^2 + 2t - 8}}$

76. $\int_{-1}^1 \frac{3 dv}{4v^2 + 4v + 4}$

78. $\int \frac{dt}{(3t+1)\sqrt{9t^2 + 6t}}$

Solving Equations with Logarithmic or Exponential Terms

In Exercises 79–84, solve for y .

79. $3^y = 2^{y+1}$

80. $4^{-y} = 3^{y+2}$

81. $9e^{2y} = x^2$

82. $3^y = 3 \ln x$

83. $\ln(y-1) = x + \ln y$

84. $\ln(10 \ln y) = \ln 5x$

Evaluating Limits

Find the limits in Exercises 85–96.

85. $\lim_{x \rightarrow 0} \frac{10^x - 1}{x}$

86. $\lim_{\theta \rightarrow 0} \frac{3^\theta - 1}{\theta}$

87. $\lim_{x \rightarrow 0} \frac{2^{\sin x} - 1}{e^x - 1}$

88. $\lim_{x \rightarrow 0} \frac{2^{-\sin x} - 1}{e^x - 1}$

89. $\lim_{x \rightarrow 0} \frac{5 - 5 \cos x}{e^x - x - 1}$

90. $\lim_{x \rightarrow 0} \frac{4 - 4e^x}{xe^x}$

91. $\lim_{t \rightarrow 0^+} \frac{t - \ln(1 + 2t)}{t^2}$

92. $\lim_{x \rightarrow 4} \frac{\sin^2(\pi x)}{e^{x-4} + 3 - x}$

93. $\lim_{t \rightarrow 0^+} \left(\frac{e^t}{t} - \frac{1}{t} \right)$

94. $\lim_{y \rightarrow 0^+} e^{-1/y} \ln y$

95. $\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x} \right)^x$

96. $\lim_{x \rightarrow 0^+} \left(1 + \frac{3}{x} \right)^x$

Comparing Growth Rates of Functions

97. Does f grow faster, slower, or at the same rate as g as $x \rightarrow \infty$? Give reasons for your answers.

a. $f(x) = \log_2 x, \quad g(x) = \log_3 x$

b. $f(x) = x, \quad g(x) = x + \frac{1}{x}$

c. $f(x) = x/100, \quad g(x) = xe^{-x}$

d. $f(x) = x, \quad g(x) = \tan^{-1} x$

e. $f(x) = \csc^{-1} x, \quad g(x) = 1/x$

f. $f(x) = \sinh x, \quad g(x) = e^x$

98. Does f grow faster, slower, or at the same rate as g as $x \rightarrow \infty$? Give reasons for your answers.

a. $f(x) = 3^{-x}, \quad g(x) = 2^{-x}$

b. $f(x) = \ln 2x, \quad g(x) = \ln x^2$

c. $f(x) = 10x^3 + 2x^2, \quad g(x) = e^x$

d. $f(x) = \tan^{-1}(1/x), \quad g(x) = 1/x$

e. $f(x) = \sin^{-1}(1/x), \quad g(x) = 1/x^2$

f. $f(x) = \operatorname{sech} x, \quad g(x) = e^{-x}$

99. True, or false? Give reasons for your answers.

- a. $\frac{1}{x^2} + \frac{1}{x^4} = O\left(\frac{1}{x^2}\right)$ b. $\frac{1}{x^2} + \frac{1}{x^4} = O\left(\frac{1}{x^4}\right)$
 c. $x = o(x + \ln x)$ d. $\ln(\ln x) = o(\ln x)$
 e. $\tan^{-1} x = O(1)$ f. $\cosh x = O(e^x)$

100. True, or false? Give reasons for your answers.

- a. $\frac{1}{x^4} = O\left(\frac{1}{x^2} + \frac{1}{x^4}\right)$ b. $\frac{1}{x^4} = o\left(\frac{1}{x^2} + \frac{1}{x^4}\right)$
 c. $\ln x = o(x + 1)$ d. $\ln 2x = O(\ln x)$
 e. $\sec^{-1} x = O(1)$ f. $\sinh x = O(e^x)$

Theory and Applications

101. The function $f(x) = e^x + x$, being differentiable and one-to-one, has a differentiable inverse $f^{-1}(x)$. Find the value of df^{-1}/dx at the point $f(\ln 2)$.

102. Find the inverse of the function $f(x) = 1 + (1/x)$, $x \neq 0$. Then show that $f^{-1}(f(x)) = f(f^{-1}(x)) = x$ and that

$$\left. \frac{df^{-1}}{dx} \right|_{f(x)} = \frac{1}{f'(x)}.$$

In Exercises 103 and 104, find the absolute maximum and minimum values of each function on the given interval.

103. $y = x \ln 2x - x$, $\left[\frac{1}{2e}, \frac{e}{2}\right]$

104. $y = 10x(2 - \ln x)$, $(0, e^2]$

105. **Area** Find the area between the curve $y = 2(\ln x)/x$ and the x -axis from $x = 1$ to $x = e$.

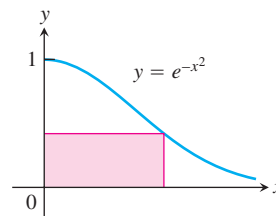
106. **Area**

- a. Show that the area between the curve $y = 1/x$ and the x -axis from $x = 10$ to $x = 20$ is the same as the area between the curve and the x -axis from $x = 1$ to $x = 2$.
 b. Show that the area between the curve $y = 1/x$ and the x -axis from ka to kb is the same as the area between the curve and the x -axis from $x = a$ to $x = b$ ($0 < a < b$, $k > 0$).

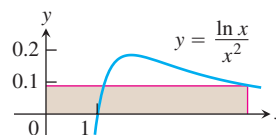
107. A particle is traveling upward and to the right along the curve $y = \ln x$. Its x -coordinate is increasing at the rate $(dx/dt) = \sqrt{x}$ m/sec. At what rate is the y -coordinate changing at the point $(e^2, 2)$?

108. A girl is sliding down a slide shaped like the curve $y = 9e^{-x/3}$. Her y -coordinate is changing at the rate $dy/dt = (-1/4)\sqrt{9 - y}$ ft/sec. At approximately what rate is her x -coordinate changing when she reaches the bottom of the slide at $x = 9$ ft? (Take e^3 to be 20 and round your answer to the nearest ft/sec.)

109. The rectangle shown here has one side on the positive y -axis, one side on the positive x -axis, and its upper right-hand vertex on the curve $y = e^{-x^2}$. What dimensions give the rectangle its largest area, and what is that area?



110. The rectangle shown here has one side on the positive y -axis, one side on the positive x -axis, and its upper right-hand vertex on the curve $y = (\ln x)/x^2$. What dimensions give the rectangle its largest area, and what is that area?



111. The functions $f(x) = \ln 5x$ and $g(x) = \ln 3x$ differ by a constant. What constant? Give reasons for your answer.

112. a. If $(\ln x)/x = (\ln 2)/2$, must $x = 2$?
 b. If $(\ln x)/x = -2 \ln 2$, must $x = 1/2$?

Give reasons for your answers.

113. The quotient $(\log_4 x)/(\log_2 x)$ has a constant value. What value? Give reasons for your answer.

T 114. **$\log_x(2)$ vs. $\log_2(x)$** How does $f(x) = \log_x(2)$ compare with $g(x) = \log_2(x)$? Here is one way to find out.

- a. Use the equation $\log_a b = (\ln b)/(\ln a)$ to express $f(x)$ and $g(x)$ in terms of natural logarithms.
 b. Graph f and g together. Comment on the behavior of f in relation to the signs and values of g .

T 115. Graph the following functions and use what you see to locate and estimate the extreme values, identify the coordinates of the inflection points, and identify the intervals on which the graphs are concave up and concave down. Then confirm your estimates by working with the functions' derivatives.

a. $y = (\ln x)/\sqrt{x}$ b. $y = e^{-x^2}$ c. $y = (1 + x)e^{-x}$

T 116. Graph $f(x) = x \ln x$. Does the function appear to have an absolute minimum value? Confirm your answer with calculus.

117. What is the age of a sample of charcoal in which 90% of the carbon-14 originally present has decayed?

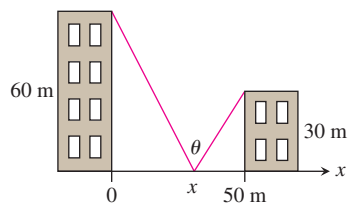
118. **Cooling a pie** A deep-dish apple pie, whose internal temperature was 220°F when removed from the oven, was set out on a breezy 40°F porch to cool. Fifteen minutes later, the pie's internal temperature was 180°F. How long did it take the pie to cool from there to 70°F?

119. **Locating a solar station** You are under contract to build a solar station at ground level on the east-west line between the two buildings shown here. How far from the taller building should you place the station to maximize the number of hours it will be

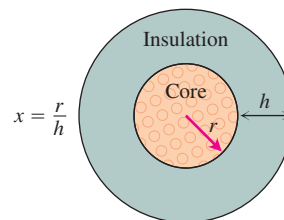
in the sun on a day when the sun passes directly overhead? Begin by observing that

$$\theta = \pi - \cot^{-1} \frac{x}{60} - \cot^{-1} \frac{50 - x}{30}.$$

Then find the value of x that maximizes θ .



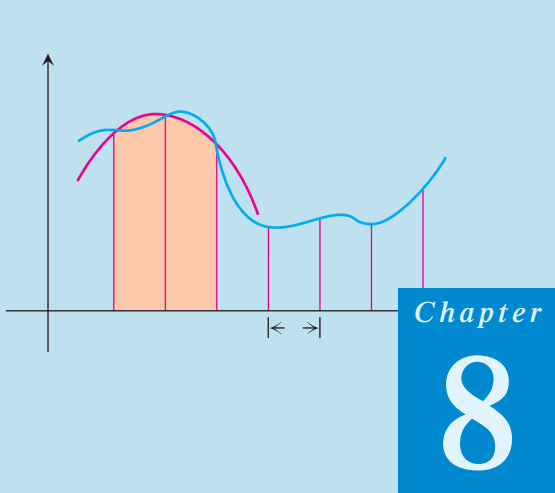
- 120.** A round underwater transmission cable consists of a core of copper wires surrounded by nonconducting insulation. If x denotes the ratio of the radius of the core to the thickness of the insulation, it is known that the speed of the transmission signal is given by the equation $v = x^2 \ln(1/x)$. If the radius of the core is 1 cm, what insulation thickness h will allow the greatest transmission speed?



Chapter 7**Questions to Guide Your Review**

1. What functions have inverses? How do you know if two functions f and g are inverses of one another? Give examples of functions that are (are not) inverses of one another.
2. How are the domains, ranges, and graphs of functions and their inverses related? Give an example.
3. How can you sometimes express the inverse of a function of x as a function of x ?
4. Under what circumstances can you be sure that the inverse of a function f is differentiable? How are the derivatives of f and f^{-1} related?

5. What is the natural logarithm function? What are its domain, range, and derivative? What arithmetic properties does it have? Comment on its graph.
6. What is logarithmic differentiation? Give an example.
7. What integrals lead to logarithms? Give examples. What are the integrals of $\tan x$ and $\cot x$?
8. How is the exponential function e^x defined? What are its domain, range, and derivative? What laws of exponents does it obey? Comment on its graph.
9. How are the functions a^x and $\log_a x$ defined? Are there any restrictions on a ? How is the graph of $\log_a x$ related to the graph of $\ln x$? What truth is there in the statement that there is really only one exponential function and one logarithmic function?
10. Describe some of the applications of base 10 logarithms.
11. What is the law of exponential change? How can it be derived from an initial value problem? What are some of the applications of the law?
12. How do you compare the growth rates of positive functions as $x \rightarrow \infty$?
13. What roles do the functions e^x and $\ln x$ play in growth comparisons?
14. Describe big-oh and little-oh notation. Give examples.
15. Which is more efficient—a sequential search or a binary search? Explain.
16. How are the inverse trigonometric functions defined? How can you sometimes use right triangles to find values of these functions? Give examples.
17. What are the derivatives of the inverse trigonometric functions? How do the domains of the derivatives compare with the domains of the functions?
18. What integrals lead to inverse trigonometric functions? How do substitution and completing the square broaden the application of these integrals?
19. What are the six basic hyperbolic functions? Comment on their domains, ranges, and graphs. What are some of the identities relating them?
20. What are the derivatives of the six basic hyperbolic functions? What are the corresponding integral formulas? What similarities do you see here with the six basic trigonometric functions?
21. How are the inverse hyperbolic functions defined? Comment on their domains, ranges, and graphs. How can you find values of $\operatorname{sech}^{-1} x$, $\operatorname{csch}^{-1} x$, and $\operatorname{coth}^{-1} x$ using a calculator's keys for $\cosh^{-1} x$, $\sinh^{-1} x$, and $\tanh^{-1} x$?
22. What integrals lead naturally to inverse hyperbolic functions?



Chapter 8

TECHNIQUES OF INTEGRATION

OVERVIEW The Fundamental Theorem connects antiderivatives and the definite integral. Evaluating the indefinite integral

$$\int f(x) dx$$

is equivalent to finding a function F such that $F'(x) = f(x)$, and then adding an arbitrary constant C :

$$\int f(x) dx = F(x) + C.$$

In this chapter we study a number of important techniques for finding indefinite integrals of more complicated functions than those seen before. The goal of this chapter is to show how to change unfamiliar integrals into integrals we can recognize, find in a table, or evaluate with a computer. We also extend the idea of the definite integral to *improper integrals* for which the integrand may be unbounded over the interval of integration, or the interval itself may no longer be finite.

8.1

Basic Integration Formulas

To help us in the search for finding indefinite integrals, it is useful to build up a table of integral formulas by inverting formulas for derivatives, as we have done in previous chapters. Then we try to match any integral that confronts us against one of the standard types. This usually involves a certain amount of algebraic manipulation as well as use of the Substitution Rule.

Recall the Substitution Rule from Section 5.5:

$$\int f(g(x))g'(x) dx = \int f(u) du$$

where $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I . Success in integration often hinges on the ability to spot what part of the integrand should be called u in order that one will also have du , so that a known formula can be applied. This means that the first requirement for skill in integration is a thorough mastery of the formulas for differentiation.

Table 8.1 shows the basic forms of integrals we have evaluated so far. In this section we present several algebraic or substitution methods to help us use this table. There is a more extensive table at the back of the book; we discuss its use in Section 8.6.

TABLE 8.1 Basic integration formulas

1. $\int du = u + C$	13. $\int \cot u \, du = \ln \sin u + C$ $= -\ln \csc u + C$
2. $\int k \, du = ku + C$ (any number k)	14. $\int e^u \, du = e^u + C$
3. $\int (du + dv) = \int du + \int dv$	15. $\int a^u \, du = \frac{a^u}{\ln a} + C$ ($a > 0, a \neq 1$)
4. $\int u^n \, du = \frac{u^{n+1}}{n+1} + C$ ($n \neq -1$)	16. $\int \sinh u \, du = \cosh u + C$
5. $\int \frac{du}{u} = \ln u + C$	17. $\int \cosh u \, du = \sinh u + C$
6. $\int \sin u \, du = -\cos u + C$	18. $\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1} \left(\frac{u}{a} \right) + C$
7. $\int \cos u \, du = \sin u + C$	19. $\int \frac{du}{a^2 + u^2} = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + C$
8. $\int \sec^2 u \, du = \tan u + C$	20. $\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a} \sec^{-1} \left \frac{u}{a} \right + C$
9. $\int \csc^2 u \, du = -\cot u + C$	21. $\int \frac{du}{\sqrt{a^2 + u^2}} = \sinh^{-1} \left(\frac{u}{a} \right) + C$ ($a > 0$)
10. $\int \sec u \tan u \, du = \sec u + C$	22. $\int \frac{du}{\sqrt{u^2 - a^2}} = \cosh^{-1} \left(\frac{u}{a} \right) + C$ ($u > a > 0$)
11. $\int \csc u \cot u \, du = -\csc u + C$	
12. $\int \tan u \, du = -\ln \cos u + C$ $= \ln \sec u + C$	

We often have to rewrite an integral to match it to a standard formula.

EXAMPLE 1 Making a Simplifying Substitution

Evaluate

$$\int \frac{2x - 9}{\sqrt{x^2 - 9x + 1}} \, dx.$$

Solution

$$\begin{aligned}
 \int \frac{2x - 9}{\sqrt{x^2 - 9x + 1}} dx &= \int \frac{du}{\sqrt{u}} && u = x^2 - 9x + 1, \\
 &= \int u^{-1/2} du && du = (2x - 9) dx. \\
 &= \frac{u^{(-1/2)+1}}{(-1/2) + 1} + C && \text{Table 8.1 Formula 4,} \\
 &= 2u^{1/2} + C && \text{with } n = -1/2 \\
 &= 2\sqrt{x^2 - 9x + 1} + C
 \end{aligned}$$

EXAMPLE 2 Completing the Square

Evaluate

$$\int \frac{dx}{\sqrt{8x - x^2}}.$$

Solution We complete the square to simplify the denominator:

$$\begin{aligned}
 8x - x^2 &= -(x^2 - 8x) = -(x^2 - 8x + 16 - 16) \\
 &= -(x^2 - 8x + 16) + 16 = 16 - (x - 4)^2.
 \end{aligned}$$

Then

$$\begin{aligned}
 \int \frac{dx}{\sqrt{8x - x^2}} &= \int \frac{dx}{\sqrt{16 - (x - 4)^2}} \\
 &= \int \frac{du}{\sqrt{a^2 - u^2}} && a = 4, u = (x - 4), \\
 &= \sin^{-1} \left(\frac{u}{a} \right) + C && du = dx \\
 &= \sin^{-1} \left(\frac{x - 4}{4} \right) + C. && \text{Table 8.1, Formula 18}
 \end{aligned}$$

EXAMPLE 3 Expanding a Power and Using a Trigonometric Identity

Evaluate

$$\int (\sec x + \tan x)^2 dx.$$

Solution We expand the integrand and get

$$(\sec x + \tan x)^2 = \sec^2 x + 2 \sec x \tan x + \tan^2 x.$$

The first two terms on the right-hand side of this equation are familiar; we can integrate them at once. How about $\tan^2 x$? There is an identity that connects it with $\sec^2 x$:

$$\tan^2 x + 1 = \sec^2 x, \quad \tan^2 x = \sec^2 x - 1.$$

We replace $\tan^2 x$ by $\sec^2 x - 1$ and get

$$\begin{aligned}\int (\sec x + \tan x)^2 dx &= \int (\sec^2 x + 2 \sec x \tan x + \sec^2 x - 1) dx \\ &= 2 \int \sec^2 x dx + 2 \int \sec x \tan x dx - \int 1 dx \\ &= 2 \tan x + 2 \sec x - x + C.\end{aligned}$$

EXAMPLE 4 Eliminating a Square Root

Evaluate

$$\int_0^{\pi/4} \sqrt{1 + \cos 4x} dx.$$

Solution We use the identity

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}, \quad \text{or} \quad 1 + \cos 2\theta = 2 \cos^2 \theta.$$

With $\theta = 2x$, this identity becomes

$$1 + \cos 4x = 2 \cos^2 2x.$$

Hence,

$$\begin{aligned}\int_0^{\pi/4} \sqrt{1 + \cos 4x} dx &= \int_0^{\pi/4} \sqrt{2} \sqrt{\cos^2 2x} dx && \sqrt{u^2} = |u| \\ &= \sqrt{2} \int_0^{\pi/4} |\cos 2x| dx && \text{On } [0, \pi/4], \cos 2x \geq 0, \\ &= \sqrt{2} \int_0^{\pi/4} \cos 2x dx && \text{so } |\cos 2x| = \cos 2x. \\ &= \sqrt{2} \left[\frac{\sin 2x}{2} \right]_0^{\pi/4} && \text{Table 8.1, Formula 7, with } u = 2x \text{ and } du = 2 dx \\ &= \sqrt{2} \left[\frac{1}{2} - 0 \right] = \frac{\sqrt{2}}{2}.\end{aligned}$$

EXAMPLE 5 Reducing an Improper Fraction

Evaluate

$$\int \frac{3x^2 - 7x}{3x + 2} dx.$$

Solution The integrand is an improper fraction (degree of numerator greater than or equal to degree of denominator). To integrate it, we divide first, getting a quotient plus a remainder that is a proper fraction:

$$\frac{3x^2 - 7x}{3x + 2} = x - 3 + \frac{6}{3x + 2}.$$

Therefore,

$$\int \frac{3x^2 - 7x}{3x + 2} dx = \int \left(x - 3 + \frac{6}{3x + 2} \right) dx = \frac{x^2}{2} - 3x + 2 \ln |3x + 2| + C. \quad \blacksquare$$

$$\begin{array}{r} \overline{) \begin{array}{r} x - 3 \\ 3x^2 - 7x \\ \underline{3x^2 + 2x} \\ -9x \\ \underline{-9x - 6} \\ + 6 \end{array}} \end{array}$$

Reducing an improper fraction by long division (Example 5) does not always lead to an expression we can integrate directly. We see what to do about that in Section 8.5.

EXAMPLE 6 Separating a Fraction

Evaluate

$$\int \frac{3x + 2}{\sqrt{1 - x^2}} dx.$$

Solution We first separate the integrand to get

$$\int \frac{3x + 2}{\sqrt{1 - x^2}} dx = 3 \int \frac{x dx}{\sqrt{1 - x^2}} + 2 \int \frac{dx}{\sqrt{1 - x^2}}.$$

In the first of these new integrals, we substitute

$$\begin{aligned} u &= 1 - x^2, & du &= -2x dx, & \text{and} & & x dx &= -\frac{1}{2} du. \\ 3 \int \frac{x dx}{\sqrt{1 - x^2}} &= 3 \int \frac{(-1/2) du}{\sqrt{u}} = -\frac{3}{2} \int u^{-1/2} du \\ &= -\frac{3}{2} \cdot \frac{u^{1/2}}{1/2} + C_1 = -3\sqrt{1 - x^2} + C_1 \end{aligned}$$

The second of the new integrals is a standard form,

$$2 \int \frac{dx}{\sqrt{1 - x^2}} = 2 \sin^{-1} x + C_2.$$

Combining these results and renaming $C_1 + C_2$ as C gives

$$\int \frac{3x + 2}{\sqrt{1 - x^2}} dx = -3\sqrt{1 - x^2} + 2 \sin^{-1} x + C. \quad \blacksquare$$

The final example of this section calculates an important integral by the algebraic technique of multiplying the integrand by a form of 1 to change the integrand into one we can integrate.

EXAMPLE 7 Integral of $y = \sec x$ —Multiplying by a Form of 1

Evaluate

$$\int \sec x dx.$$

Solution

$$\begin{aligned} \int \sec x dx &= \int (\sec x)(1) dx = \int \sec x \cdot \frac{\sec x + \tan x}{\sec x + \tan x} dx \\ &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx \\ &= \int \frac{du}{u} \\ &= \ln |u| + C = \ln |\sec x + \tan x| + C. \end{aligned}$$

$$\begin{aligned} u &= \tan x + \sec x, \\ du &= (\sec^2 x + \sec x \tan x) dx \end{aligned}$$

HISTORICAL BIOGRAPHY

George David Birkhoff
(1884–1944)

With cosecants and cotangents in place of secants and tangents, the method of Example 7 leads to a companion formula for the integral of the cosecant (see Exercise 95).

TABLE 8.2 The secant and cosecant integrals

1. $\int \sec u \, du = \ln |\sec u + \tan u| + C$
2. $\int \csc u \, du = -\ln |\csc u + \cot u| + C$

Procedures for Matching Integrals to Basic Formulas

PROCEDURE	EXAMPLE
Making a simplifying substitution	$\frac{2x - 9}{\sqrt{x^2 - 9x + 1}} \, dx = \frac{du}{\sqrt{u}}$
Completing the square	$\sqrt{8x - x^2} = \sqrt{16 - (x - 4)^2}$
Using a trigonometric identity	$\begin{aligned} (\sec x + \tan x)^2 &= \sec^2 x + 2 \sec x \tan x + \tan^2 x \\ &= \sec^2 x + 2 \sec x \tan x \\ &\quad + (\sec^2 x - 1) \\ &= 2 \sec^2 x + 2 \sec x \tan x - 1 \end{aligned}$
Eliminating a square root	$\sqrt{1 + \cos 4x} = \sqrt{2 \cos^2 2x} = \sqrt{2} \cos 2x $
Reducing an improper fraction	$\frac{3x^2 - 7x}{3x + 2} = x - 3 + \frac{6}{3x + 2}$
Separating a fraction	$\frac{3x + 2}{\sqrt{1 - x^2}} = \frac{3x}{\sqrt{1 - x^2}} + \frac{2}{\sqrt{1 - x^2}}$
Multiplying by a form of 1	$\begin{aligned} \sec x &= \sec x \cdot \frac{\sec x + \tan x}{\sec x + \tan x} \\ &= \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \end{aligned}$

EXERCISES 8.1

Basic Substitutions

Evaluate each integral in Exercises 1–36 by using a substitution to reduce it to standard form.

1. $\int \frac{16x \, dx}{\sqrt{8x^2 + 1}}$

2. $\int \frac{3 \cos x \, dx}{\sqrt{1 + 3 \sin x}}$

3. $\int 3\sqrt{\sin v} \cos v \, dv$

5. $\int_0^1 \frac{16x \, dx}{8x^2 + 2}$

4. $\int \cot^3 y \csc^2 y \, dy$

6. $\int_{\pi/4}^{\pi/3} \frac{\sec^2 z}{\tan z} \, dz$

7. $\int \frac{dx}{\sqrt{x}(\sqrt{x} + 1)}$
8. $\int \frac{dx}{x - \sqrt{x}}$
9. $\int \cot(3 - 7x) dx$
10. $\int \csc(\pi x - 1) dx$
11. $\int e^\theta \csc(e^\theta + 1) d\theta$
12. $\int \frac{\cot(3 + \ln x)}{x} dx$
13. $\int \sec \frac{t}{3} dt$
14. $\int x \sec(x^2 - 5) dx$
15. $\int \csc(s - \pi) ds$
16. $\int \frac{1}{\theta^2} \csc \frac{1}{\theta} d\theta$
17. $\int_0^{\sqrt{\ln 2}} 2x e^{x^2} dx$
18. $\int_{\pi/2}^{\pi} (\sin y) e^{\cos y} dy$
19. $\int e^{\tan v} \sec^2 v dv$
20. $\int \frac{e^{\sqrt{t}} dt}{\sqrt{t}}$
21. $\int 3^{x+1} dx$
22. $\int \frac{2^{\ln x}}{x} dx$
23. $\int \frac{2^{\sqrt{w}} dw}{2\sqrt{w}}$
24. $\int 10^{2\theta} d\theta$
25. $\int \frac{9 du}{1 + 9u^2}$
26. $\int \frac{4 dx}{1 + (2x + 1)^2}$
27. $\int_0^{1/6} \frac{dx}{\sqrt{1 - 9x^2}}$
28. $\int_0^1 \frac{dt}{\sqrt{4 - t^2}}$
29. $\int \frac{2s ds}{\sqrt{1 - s^4}}$
30. $\int \frac{2 dx}{x\sqrt{1 - 4 \ln^2 x}}$
31. $\int \frac{6 dx}{x\sqrt{25x^2 - 1}}$
32. $\int \frac{dr}{r\sqrt{r^2 - 9}}$
33. $\int \frac{dx}{e^x + e^{-x}}$
34. $\int \frac{dy}{\sqrt{e^{2y} - 1}}$
35. $\int_1^{e^{\pi/3}} \frac{dx}{x \cos(\ln x)}$
36. $\int \frac{\ln x dx}{x + 4x \ln^2 x}$

Completing the Square

Evaluate each integral in Exercises 37–42 by completing the square and using a substitution to reduce it to standard form.

37. $\int_1^2 \frac{8 dx}{x^2 - 2x + 2}$
38. $\int_2^4 \frac{2 dx}{x^2 - 6x + 10}$
39. $\int \frac{dt}{\sqrt{-t^2 + 4t - 3}}$
40. $\int \frac{d\theta}{\sqrt{2\theta - \theta^2}}$
41. $\int \frac{dx}{(x + 1)\sqrt{x^2 + 2x}}$
42. $\int \frac{dx}{(x - 2)\sqrt{x^2 - 4x + 3}}$

Trigonometric Identities

Evaluate each integral in Exercises 43–46 by using trigonometric identities and substitutions to reduce it to standard form.

43. $\int (\sec x + \cot x)^2 dx$
44. $\int (\csc x - \tan x)^2 dx$
45. $\int \csc x \sin 3x dx$
46. $\int (\sin 3x \cos 2x - \cos 3x \sin 2x) dx$

Improper Fractions

Evaluate each integral in Exercises 47–52 by reducing the improper fraction and using a substitution (if necessary) to reduce it to standard form.

47. $\int \frac{x}{x + 1} dx$
48. $\int \frac{x^2}{x^2 + 1} dx$
49. $\int_{\sqrt{2}}^3 \frac{2x^3}{x^2 - 1} dx$
50. $\int_{-1}^3 \frac{4x^2 - 7}{2x + 3} dx$
51. $\int \frac{4t^3 - t^2 + 16t}{t^2 + 4} dt$
52. $\int \frac{2\theta^3 - 7\theta^2 + 7\theta}{2\theta - 5} d\theta$

Separating Fractions

Evaluate each integral in Exercises 53–56 by separating the fraction and using a substitution (if necessary) to reduce it to standard form.

53. $\int \frac{1 - x}{\sqrt{1 - x^2}} dx$
54. $\int \frac{x + 2\sqrt{x - 1}}{2x\sqrt{x - 1}} dx$
55. $\int_0^{\pi/4} \frac{1 + \sin x}{\cos^2 x} dx$
56. $\int_0^{1/2} \frac{2 - 8x}{1 + 4x^2} dx$

Multiplying by a Form of 1

Evaluate each integral in Exercises 57–62 by multiplying by a form of 1 and using a substitution (if necessary) to reduce it to standard form.

57. $\int \frac{1}{1 + \sin x} dx$
58. $\int \frac{1}{1 + \cos x} dx$
59. $\int \frac{1}{\sec \theta + \tan \theta} d\theta$
60. $\int \frac{1}{\csc \theta + \cot \theta} d\theta$
61. $\int \frac{1}{1 - \sec x} dx$
62. $\int \frac{1}{1 - \csc x} dx$

Eliminating Square Roots

Evaluate each integral in Exercises 63–70 by eliminating the square root.

63. $\int_0^{2\pi} \sqrt{\frac{1 - \cos x}{2}} dx$
64. $\int_0^{\pi} \sqrt{1 - \cos 2x} dx$

$$\begin{array}{ll}
65. \int_{\pi/2}^{\pi} \sqrt{1 + \cos 2t} \, dt & 66. \int_{-\pi}^0 \sqrt{1 + \cos t} \, dt \\
67. \int_{-\pi}^0 \sqrt{1 - \cos^2 \theta} \, d\theta & 68. \int_{\pi/2}^{\pi} \sqrt{1 - \sin^2 \theta} \, d\theta \\
69. \int_{-\pi/4}^{\pi/4} \sqrt{1 + \tan^2 y} \, dy & 70. \int_{-\pi/4}^0 \sqrt{\sec^2 y - 1} \, dy
\end{array}$$

Assorted Integrations

Evaluate each integral in Exercises 71–82 by using any technique you think is appropriate.

$$\begin{array}{ll}
71. \int_{\pi/4}^{3\pi/4} (\csc x - \cot x)^2 \, dx & 72. \int_0^{\pi/4} (\sec x + 4 \cos x)^2 \, dx \\
73. \int \cos \theta \csc(\sin \theta) \, d\theta & 74. \int \left(1 + \frac{1}{x}\right) \cot(x + \ln x) \, dx \\
75. \int (\csc x - \sec x)(\sin x + \cos x) \, dx & \\
76. \int 3 \sinh\left(\frac{x}{2} + \ln 5\right) \, dx & \\
77. \int \frac{6 \, dy}{\sqrt{y}(1+y)} & 78. \int \frac{dx}{x\sqrt{4x^2 - 1}} \\
79. \int \frac{7 \, dx}{(x-1)\sqrt{x^2 - 2x - 48}} & 80. \int \frac{dx}{(2x+1)\sqrt{4x^2 + 4x}} \\
81. \int \sec^2 t \tan(\tan t) \, dt & 82. \int \frac{dx}{x\sqrt{3+x^2}}
\end{array}$$

Trigonometric Powers

83. a. Evaluate $\int \cos^3 \theta \, d\theta$. (Hint: $\cos^2 \theta = 1 - \sin^2 \theta$.)
 b. Evaluate $\int \cos^5 \theta \, d\theta$.
 c. Without actually evaluating the integral, explain how you would evaluate $\int \cos^9 \theta \, d\theta$.
84. a. Evaluate $\int \sin^3 \theta \, d\theta$. (Hint: $\sin^2 \theta = 1 - \cos^2 \theta$.)
 b. Evaluate $\int \sin^5 \theta \, d\theta$.
 c. Evaluate $\int \sin^7 \theta \, d\theta$.
 d. Without actually evaluating the integral, explain how you would evaluate $\int \sin^{13} \theta \, d\theta$.
85. a. Express $\int \tan^3 \theta \, d\theta$ in terms of $\int \tan \theta \, d\theta$. Then evaluate $\int \tan^3 \theta \, d\theta$. (Hint: $\tan^2 \theta = \sec^2 \theta - 1$.)
 b. Express $\int \tan^5 \theta \, d\theta$ in terms of $\int \tan^3 \theta \, d\theta$.
 c. Express $\int \tan^7 \theta \, d\theta$ in terms of $\int \tan^5 \theta \, d\theta$.
 d. Express $\int \tan^{2k+1} \theta \, d\theta$, where k is a positive integer, in terms of $\int \tan^{2k-1} \theta \, d\theta$.
86. a. Express $\int \cot^3 \theta \, d\theta$ in terms of $\int \cot \theta \, d\theta$. Then evaluate $\int \cot^3 \theta \, d\theta$. (Hint: $\cot^2 \theta = \csc^2 \theta - 1$.)

- b. Express $\int \cot^5 \theta \, d\theta$ in terms of $\int \cot^3 \theta \, d\theta$.
 c. Express $\int \cot^7 \theta \, d\theta$ in terms of $\int \cot^5 \theta \, d\theta$.
 d. Express $\int \cot^{2k+1} \theta \, d\theta$, where k is a positive integer, in terms of $\int \cot^{2k-1} \theta \, d\theta$.

Theory and Examples

87. **Area** Find the area of the region bounded above by $y = 2 \cos x$ and below by $y = \sec x$, $-\pi/4 \leq x \leq \pi/4$.
88. **Area** Find the area of the “triangular” region that is bounded from above and below by the curves $y = \csc x$ and $y = \sin x$, $\pi/6 \leq x \leq \pi/2$, and on the left by the line $x = \pi/6$.
89. **Volume** Find the volume of the solid generated by revolving the region in Exercise 87 about the x -axis.
90. **Volume** Find the volume of the solid generated by revolving the region in Exercise 88 about the x -axis.
91. **Arc length** Find the length of the curve $y = \ln(\cos x)$, $0 \leq x \leq \pi/3$.
92. **Arc length** Find the length of the curve $y = \ln(\sec x)$, $0 \leq x \leq \pi/4$.
93. **Centroid** Find the centroid of the region bounded by the x -axis, the curve $y = \sec x$, and the lines $x = -\pi/4$, $x = \pi/4$.
94. **Centroid** Find the centroid of the region that is bounded by the x -axis, the curve $y = \csc x$, and the lines $x = \pi/6$, $x = 5\pi/6$.
95. **The integral of $\csc x$** Repeat the derivation in Example 7, using cofunctions, to show that

$$\int \csc x \, dx = -\ln |\csc x + \cot x| + C.$$

96. **Using different substitutions** Show that the integral

$$\int ((x^2 - 1)(x + 1))^{-2/3} \, dx$$

can be evaluated with any of the following substitutions.

- a. $u = 1/(x + 1)$
 b. $u = ((x - 1)/(x + 1))^k$ for $k = 1, 1/2, 1/3, -1/3, -2/3$, and -1
 c. $u = \tan^{-1} x$
 d. $u = \tan^{-1} \sqrt{x}$ e. $u = \tan^{-1} ((x - 1)/2)$
 f. $u = \cos^{-1} x$ g. $u = \cosh^{-1} x$

What is the value of the integral? (Source: “Problems and Solutions,” *College Mathematics Journal*, Vol. 21, No. 5 (Nov. 1990), pp. 425–426.)

8.2

Integration by Parts

Since

$$\int x \, dx = \frac{1}{2}x^2 + C$$

and

$$\int x^2 \, dx = \frac{1}{3}x^3 + C,$$

it is apparent that

$$\int x \cdot x \, dx \neq \int x \, dx \cdot \int x \, dx.$$

In other words, the integral of a product is generally *not* the product of the individual-integrals:

$$\int f(x)g(x) \, dx \text{ is not equal to } \int f(x) \, dx \cdot \int g(x) \, dx.$$

Integration by parts is a technique for simplifying integrals of the form

$$\int f(x)g(x) \, dx.$$

It is useful when f can be differentiated repeatedly and g can be integrated repeatedly without difficulty. The integral

$$\int xe^x \, dx$$

is such an integral because $f(x) = x$ can be differentiated twice to become zero and $g(x) = e^x$ can be integrated repeatedly without difficulty. Integration by parts also applies to integrals like

$$\int e^x \sin x \, dx$$

in which each part of the integrand appears again after repeated differentiation or integration.

In this section, we describe integration by parts and show how to apply it.

Product Rule in Integral Form

If f and g are differentiable functions of x , the Product Rule says

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x).$$

In terms of indefinite integrals, this equation becomes

$$\int \frac{d}{dx}[f(x)g(x)] \, dx = \int [f'(x)g(x) + f(x)g'(x)] \, dx$$

or

$$\int \frac{d}{dx} [f(x)g(x)] dx = \int f'(x)g(x) dx + \int f(x)g'(x) dx.$$

Rearranging the terms of this last equation, we get

$$\int f(x)g'(x) dx = \int \frac{d}{dx} [f(x)g(x)] dx - \int f'(x)g(x) dx$$

leading to the **integration by parts** formula

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx \quad (1)$$

Sometimes it is easier to remember the formula if we write it in differential form. Let $u = f(x)$ and $v = g(x)$. Then $du = f'(x) dx$ and $dv = g'(x) dx$. Using the Substitution Rule, the integration by parts formula becomes

Integration by Parts Formula

$$\int u dv = uv - \int v du \quad (2)$$

This formula expresses one integral, $\int u dv$, in terms of a second integral, $\int v du$. With a proper choice of u and v , the second integral may be easier to evaluate than the first. In using the formula, various choices may be available for u and dv . The next examples illustrate the technique.

EXAMPLE 1 Using Integration by Parts

Find

$$\int x \cos x dx.$$

Solution We use the formula $\int u dv = uv - \int v du$ with

$$\begin{aligned} u &= x, & dv &= \cos x dx, \\ du &= dx, & v &= \sin x. \end{aligned} \quad \text{Simplest antiderivative of } \cos x$$

Then

$$\int x \cos x dx = x \sin x - \int \sin x dx = x \sin x + \cos x + C. \quad \blacksquare$$

Let us examine the choices available for u and dv in Example 1.

EXAMPLE 2 Example 1 Revisited

To apply integration by parts to

$$\int x \cos x dx = \int u dv$$

we have four possible choices:

1. Let $u = 1$ and $dv = x \cos x \, dx$.
2. Let $u = x$ and $dv = \cos x \, dx$.
3. Let $u = x \cos x$ and $dv = dx$.
4. Let $u = \cos x$ and $dv = x \, dx$.

Let's examine these one at a time.

Choice 1 won't do because we don't know how to integrate $dv = x \cos x \, dx$ to get v .

Choice 2 works well, as we saw in Example 1.

Choice 3 leads to

$$\begin{aligned} u &= x \cos x, & dv &= dx, \\ du &= (\cos x - x \sin x) \, dx, & v &= x, \end{aligned}$$

and the new integral

$$\int v \, du = \int (x \cos x - x^2 \sin x) \, dx.$$

This is worse than the integral we started with.

Choice 4 leads to

$$\begin{aligned} u &= \cos x, & dv &= x \, dx, \\ du &= -\sin x \, dx, & v &= x^2/2, \end{aligned}$$

so the new integral is

$$\int v \, du = -\int \frac{x^2}{2} \sin x \, dx.$$

This, too, is worse. ■

The goal of integration by parts is to go from an integral $\int u \, dv$ that we don't see how to evaluate to an integral $\int v \, du$ that we can evaluate. Generally, you choose dv first to be as much of the integrand, including dx , as you can readily integrate; u is the leftover part. Keep in mind that integration by parts does not always work.

EXAMPLE 3 Integral of the Natural Logarithm

Find

$$\int \ln x \, dx.$$

Solution Since $\int \ln x \, dx$ can be written as $\int \ln x \cdot 1 \, dx$, we use the formula $\int u \, dv = uv - \int v \, du$ with

$$\begin{aligned} u &= \ln x && \text{Simplifies when differentiated} && dv &= dx && \text{Easy to integrate} \\ du &= \frac{1}{x} \, dx, && && v &= x. && \text{Simplest antiderivative} \end{aligned}$$

Then

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx = x \ln x - \int dx = x \ln x - x + C. \quad \text{■}$$

Sometimes we have to use integration by parts more than once.

EXAMPLE 4 Repeated Use of Integration by Parts

Evaluate

$$\int x^2 e^x dx.$$

Solution With $u = x^2$, $dv = e^x dx$, $du = 2x dx$, and $v = e^x$, we have

$$\int x^2 e^x dx = x^2 e^x - 2 \int x e^x dx.$$

The new integral is less complicated than the original because the exponent on x is reduced by one. To evaluate the integral on the right, we integrate by parts again with $u = x$, $dv = e^x dx$. Then $du = dx$, $v = e^x$, and

$$\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C.$$

Hence,

$$\begin{aligned} \int x^2 e^x dx &= x^2 e^x - 2 \int x e^x dx \\ &= x^2 e^x - 2x e^x + 2e^x + C. \end{aligned}$$

The technique of Example 4 works for any integral $\int x^n e^x dx$ in which n is a positive integer, because differentiating x^n will eventually lead to zero and integrating e^x is easy. We say more about this later in this section when we discuss *tabular integration*.

Integrals like the one in the next example occur in electrical engineering. Their evaluation requires two integrations by parts, followed by solving for the unknown integral.

EXAMPLE 5 Solving for the Unknown Integral

Evaluate

$$\int e^x \cos x dx.$$

Solution Let $u = e^x$ and $dv = \cos x dx$. Then $du = e^x dx$, $v = \sin x$, and

$$\int e^x \cos x dx = e^x \sin x - \int e^x \sin x dx.$$

The second integral is like the first except that it has $\sin x$ in place of $\cos x$. To evaluate it, we use integration by parts with

$$u = e^x, \quad dv = \sin x dx, \quad v = -\cos x, \quad du = e^x dx.$$

Then

$$\begin{aligned} \int e^x \cos x dx &= e^x \sin x - \left(-e^x \cos x - \int (-\cos x)(e^x dx) \right) \\ &= e^x \sin x + e^x \cos x - \int e^x \cos x dx. \end{aligned}$$

The unknown integral now appears on both sides of the equation. Adding the integral to both sides and adding the constant of integration gives

$$2 \int e^x \cos x \, dx = e^x \sin x + e^x \cos x + C_1.$$

Dividing by 2 and renaming the constant of integration gives

$$\int e^x \cos x \, dx = \frac{e^x \sin x + e^x \cos x}{2} + C. \quad \blacksquare$$

Evaluating Definite Integrals by Parts

The integration by parts formula in Equation (1) can be combined with Part 2 of the Fundamental Theorem in order to evaluate definite integrals by parts. Assuming that both f' and g' are continuous over the interval $[a, b]$, Part 2 of the Fundamental Theorem gives

Integration by Parts Formula for Definite Integrals

$$\int_a^b f(x)g'(x) \, dx = f(x)g(x) \Big|_a^b - \int_a^b f'(x)g(x) \, dx \quad (3)$$

In applying Equation (3), we normally use the u and v notation from Equation (2) because it is easier to remember. Here is an example.

EXAMPLE 6 Finding Area

Find the area of the region bounded by the curve $y = xe^{-x}$ and the x -axis from $x = 0$ to $x = 4$.

Solution The region is shaded in Figure 8.1. Its area is

$$\int_0^4 xe^{-x} \, dx.$$

Let $u = x$, $dv = e^{-x} \, dx$, $v = -e^{-x}$, and $du = dx$. Then,

$$\begin{aligned} \int_0^4 xe^{-x} \, dx &= -xe^{-x} \Big|_0^4 - \int_0^4 (-e^{-x}) \, dx \\ &= [-4e^{-4} - (0)] + \int_0^4 e^{-x} \, dx \\ &= -4e^{-4} - e^{-x} \Big|_0^4 \\ &= -4e^{-4} - e^{-4} - (-e^0) = 1 - 5e^{-4} \approx 0.91. \quad \blacksquare \end{aligned}$$

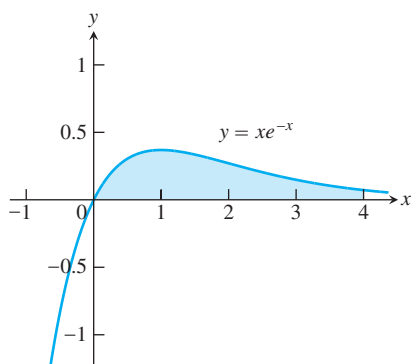


FIGURE 8.1 The region in Example 6.

Tabular Integration

We have seen that integrals of the form $\int f(x)g(x) \, dx$, in which f can be differentiated repeatedly to become zero and g can be integrated repeatedly without difficulty, are natural candidates for integration by parts. However, if many repetitions are required, the calculations can be cumbersome. In situations like this, there is a way to organize

the calculations that saves a great deal of work. It is called **tabular integration** and is illustrated in the following examples.

EXAMPLE 7 Using Tabular Integration

Evaluate

$$\int x^2 e^x \, dx.$$

Solution With $f(x) = x^2$ and $g(x) = e^x$, we list:

$f(x)$ and its derivatives		$g(x)$ and its integrals
x^2	$(+)$	e^x
$2x$	$(-)$	e^x
2	$(+)$	e^x
0		e^x

We combine the products of the functions connected by the arrows according to the operation signs above the arrows to obtain

$$\int x^2 e^x \, dx = x^2 e^x - 2x e^x + 2e^x + C.$$

Compare this with the result in Example 4. ■

EXAMPLE 8 Using Tabular Integration

Evaluate

$$\int x^3 \sin x \, dx.$$

Solution With $f(x) = x^3$ and $g(x) = \sin x$, we list:

$f(x)$ and its derivatives		$g(x)$ and its integrals
x^3	$(+)$	$\sin x$
$3x^2$	$(-)$	$-\cos x$
$6x$	$(+)$	$-\sin x$
6	$(-)$	$\cos x$
0		$\sin x$

Again we combine the products of the functions connected by the arrows according to the operation signs above the arrows to obtain

$$\int x^3 \sin x \, dx = -x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x + C. \quad \blacksquare$$

The Additional Exercises at the end of this chapter show how tabular integration can be used when neither function f nor g can be differentiated repeatedly to become zero.

Summary

When substitution doesn't work, try integration by parts. Start with an integral in which the integrand is the product of two functions,

$$\int f(x)g(x) dx.$$

(Remember that g may be the constant function 1, as in Example 3.) Match the integral with the form

$$\int u dv$$

by choosing dv to be part of the integrand including dx and either $f(x)$ or $g(x)$. Remember that we must be able to readily integrate dv to get v in order to obtain the right side of the formula

$$\int u dv = uv - \int v du.$$

If the new integral on the right side is more complex than the original one, try a different choice for u and dv .

EXAMPLE 9 A Reduction Formula

Obtain a “reduction” formula that expresses the integral

$$\int \cos^n x dx$$

in terms of an integral of a lower power of $\cos x$.

Solution We may think of $\cos^n x$ as $\cos^{n-1} x \cdot \cos x$. Then we let

$$u = \cos^{n-1} x \quad \text{and} \quad dv = \cos x dx,$$

so that

$$du = (n-1)\cos^{n-2} x (-\sin x dx) \quad \text{and} \quad v = \sin x.$$

Hence

$$\begin{aligned} \int \cos^n x dx &= \cos^{n-1} x \sin x + (n-1) \int \sin^2 x \cos^{n-2} x dx \\ &= \cos^{n-1} x \sin x + (n-1) \int (1 - \cos^2 x) \cos^{n-2} x dx, \\ &= \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x dx - (n-1) \int \cos^n x dx. \end{aligned}$$

If we add

$$(n-1) \int \cos^n x dx$$

to both sides of this equation, we obtain

$$n \int \cos^n x \, dx = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \, dx.$$

We then divide through by n , and the final result is

$$\int \cos^n x \, dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx.$$

This allows us to reduce the exponent on $\cos x$ by 2 and is a very useful formula. When n is a positive integer, we may apply the formula repeatedly until the remaining integral is either

$$\int \cos x \, dx = \sin x + C \quad \text{or} \quad \int \cos^0 x \, dx = \int dx = x + C. \quad \blacksquare$$

EXAMPLE 10 Using a Reduction Formula

Evaluate

$$\int \cos^3 x \, dx.$$

Solution From the result in Example 9,

$$\begin{aligned} \int \cos^3 x \, dx &= \frac{\cos^2 x \sin x}{3} + \frac{2}{3} \int \cos x \, dx \\ &= \frac{1}{3} \cos^2 x \sin x + \frac{2}{3} \sin x + C. \end{aligned} \quad \blacksquare$$

EXERCISES 8.2

Integration by Parts

Evaluate the integrals in Exercises 1–24.

1. $\int x \sin \frac{x}{2} dx$

2. $\int \theta \cos \pi \theta d\theta$

3. $\int t^2 \cos t dt$

4. $\int x^2 \sin x dx$

5. $\int_1^2 x \ln x dx$

6. $\int_1^e x^3 \ln x dx$

7. $\int \tan^{-1} y dy$

8. $\int \sin^{-1} y dy$

9. $\int x \sec^2 x dx$

10. $\int 4x \sec^2 2x dx$

11. $\int x^3 e^x dx$

12. $\int p^4 e^{-p} dp$

13. $\int (x^2 - 5x)e^x dx$

15. $\int x^5 e^x dx$

17. $\int_0^{\pi/2} \theta^2 \sin 2\theta d\theta$

19. $\int_{2/\sqrt{3}}^2 t \sec^{-1} t dt$

21. $\int e^\theta \sin \theta d\theta$

23. $\int e^{2x} \cos 3x dx$

14. $\int (r^2 + r + 1)e^r dr$

16. $\int t^2 e^{4t} dt$

18. $\int_0^{\pi/2} x^3 \cos 2x dx$

20. $\int_0^{1/\sqrt{2}} 2x \sin^{-1}(x^2) dx$

22. $\int e^{-y} \cos y dy$

24. $\int e^{-2x} \sin 2x dx$

Substitution and Integration by Parts

Evaluate the integrals in Exercises 25–30 by using a substitution prior to integration by parts.

25. $\int e^{\sqrt{3s+9}} ds$

26. $\int_0^1 x\sqrt{1-x} dx$

27. $\int_0^{\pi/3} x \tan^2 x dx$

28. $\int \ln(x + x^2) dx$

29. $\int \sin(\ln x) dx$

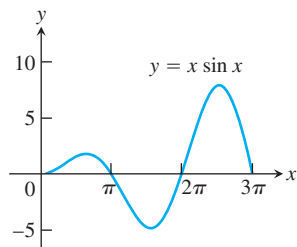
30. $\int z(\ln z)^2 dz$

Theory and Examples

31. Finding area Find the area of the region enclosed by the curve $y = x \sin x$ and the x -axis (see the accompanying figure) for

a. $0 \leq x \leq \pi$ b. $\pi \leq x \leq 2\pi$ c. $2\pi \leq x \leq 3\pi$.

d. What pattern do you see here? What is the area between the curve and the x -axis for $n\pi \leq x \leq (n+1)\pi$, n an arbitrary nonnegative integer? Give reasons for your answer.



32. Finding area Find the area of the region enclosed by the curve $y = x \cos x$ and the x -axis (see the accompanying figure) for

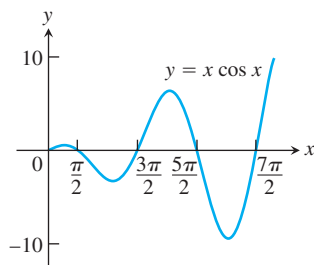
a. $\pi/2 \leq x \leq 3\pi/2$ b. $3\pi/2 \leq x \leq 5\pi/2$

c. $5\pi/2 \leq x \leq 7\pi/2$.

d. What pattern do you see? What is the area between the curve and the x -axis for

$$\left(\frac{2n-1}{2}\right)\pi \leq x \leq \left(\frac{2n+1}{2}\right)\pi,$$

n an arbitrary positive integer? Give reasons for your answer.



33. Finding volume Find the volume of the solid generated by revolving the region in the first quadrant bounded by the coordinate axes, the curve $y = e^x$, and the line $x = \ln 2$ about the line $x = \ln 2$.

34. Finding volume Find the volume of the solid generated by revolving the region in the first quadrant bounded by the coordinate axes, the curve $y = e^{-x}$, and the line $x = 1$

a. about the y -axis. b. about the line $x = 1$.

35. Finding volume Find the volume of the solid generated by revolving the region in the first quadrant bounded by the coordinate axes and the curve $y = \cos x$, $0 \leq x \leq \pi/2$, about

a. the y -axis. b. the line $x = \pi/2$.

36. Finding volume Find the volume of the solid generated by revolving the region bounded by the x -axis and the curve $y = x \sin x$, $0 \leq x \leq \pi$, about

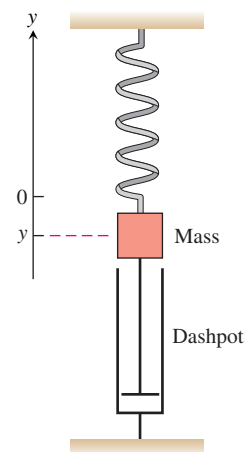
a. the y -axis. b. the line $x = \pi$.

(See Exercise 31 for a graph.)

37. Average value A retarding force, symbolized by the dashpot in the figure, slows the motion of the weighted spring so that the mass's position at time t is

$$y = 2e^{-t} \cos t, \quad t \geq 0.$$

Find the average value of y over the interval $0 \leq t \leq 2\pi$.



38. Average value In a mass-spring-dashpot system like the one in Exercise 37, the mass's position at time t is

$$y = 4e^{-t}(\sin t - \cos t), \quad t \geq 0.$$

Find the average value of y over the interval $0 \leq t \leq 2\pi$.

Reduction Formulas

In Exercises 39–42, use integration by parts to establish the *reduction formula*.

$$39. \int x^n \cos x \, dx = x^n \sin x - n \int x^{n-1} \sin x \, dx$$

$$40. \int x^n \sin x \, dx = -x^n \cos x + n \int x^{n-1} \cos x \, dx$$

$$41. \int x^n e^{ax} \, dx = \frac{x^n e^{ax}}{a} - \frac{n}{a} \int x^{n-1} e^{ax} \, dx, \quad a \neq 0$$

$$42. \int (\ln x)^n \, dx = x(\ln x)^n - n \int (\ln x)^{n-1} \, dx$$

Integrating Inverses of Functions

Integration by parts leads to a rule for integrating inverses that usually gives good results:

$$\begin{aligned} \int f^{-1}(x) \, dx &= \int y f'(y) \, dy && y = f^{-1}(x), \quad x = f(y) \\ &&& dx = f'(y) \, dy \\ &= y f(y) - \int f(y) \, dy && \text{Integration by parts with} \\ &&& u = y, \, dv = f'(y) \, dy \\ &= x f^{-1}(x) - \int f(y) \, dy \end{aligned}$$

The idea is to take the most complicated part of the integral, in this case $f^{-1}(x)$, and simplify it first. For the integral of $\ln x$, we get

$$\begin{aligned} \int \ln x \, dx &= \int y e^y \, dy && y = \ln x, \quad x = e^y \\ &&& dx = e^y \, dy \\ &= y e^y - e^y + C \\ &= x \ln x - x + C. \end{aligned}$$

For the integral of $\cos^{-1} x$ we get

$$\begin{aligned} \int \cos^{-1} x \, dx &= x \cos^{-1} x - \int \cos y \, dy && y = \cos^{-1} x \\ &= x \cos^{-1} x - \sin y + C \\ &= x \cos^{-1} x - \sin(\cos^{-1} x) + C. \end{aligned}$$

Use the formula

$$\int f^{-1}(x) \, dx = x f^{-1}(x) - \int f(y) \, dy \quad y = f^{-1}(x) \quad (4)$$

to evaluate the integrals in Exercises 43–46. Express your answers in terms of x .

$$43. \int \sin^{-1} x \, dx$$

$$44. \int \tan^{-1} x \, dx$$

$$45. \int \sec^{-1} x \, dx$$

$$46. \int \log_2 x \, dx$$

Another way to integrate $f^{-1}(x)$ (when f^{-1} is integrable, of course) is to use integration by parts with $u = f^{-1}(x)$ and $dv = dx$ to rewrite the integral of f^{-1} as

$$\int f^{-1}(x) \, dx = x f^{-1}(x) - \int x \left(\frac{d}{dx} f^{-1}(x) \right) dx. \quad (5)$$

Exercises 47 and 48 compare the results of using Equations (4) and (5).

47. Equations (4) and (5) give different formulas for the integral of $\cos^{-1} x$:

$$\text{a. } \int \cos^{-1} x \, dx = x \cos^{-1} x - \sin(\cos^{-1} x) + C \quad \text{Eq. (4)}$$

$$\text{b. } \int \cos^{-1} x \, dx = x \cos^{-1} x - \sqrt{1 - x^2} + C \quad \text{Eq. (5)}$$

Can both integrations be correct? Explain.

48. Equations (4) and (5) lead to different formulas for the integral of $\tan^{-1} x$:

$$\text{a. } \int \tan^{-1} x \, dx = x \tan^{-1} x - \ln \sec(\tan^{-1} x) + C \quad \text{Eq. (4)}$$

$$\text{b. } \int \tan^{-1} x \, dx = x \tan^{-1} x - \ln \sqrt{1 + x^2} + C \quad \text{Eq. (5)}$$

Can both integrations be correct? Explain.

Evaluate the integrals in Exercises 49 and 50 with (a) Eq. (4) and (b) Eq. (5). In each case, check your work by differentiating your answer with respect to x .

$$49. \int \sinh^{-1} x \, dx$$

$$50. \int \tanh^{-1} x \, dx$$

8.3

Integration of Rational Functions by Partial Fractions

This section shows how to express a rational function (a quotient of polynomials) as a sum of simpler fractions, called *partial fractions*, which are easily integrated. For instance, the rational function $(5x - 3)/(x^2 - 2x - 3)$ can be rewritten as

$$\frac{5x - 3}{x^2 - 2x - 3} = \frac{2}{x + 1} + \frac{3}{x - 3},$$

which can be verified algebraically by placing the fractions on the right side over a common denominator $(x + 1)(x - 3)$. The skill acquired in writing rational functions as such a sum is useful in other settings as well (for instance, when using certain transform methods to solve differential equations). To integrate the rational function $(5x - 3)/(x + 1)(x - 3)$ on the left side of our previous expression, we simply sum the integrals of the fractions on the right side:

$$\begin{aligned}\int \frac{5x - 3}{(x + 1)(x - 3)} dx &= \int \frac{2}{x + 1} dx + \int \frac{3}{x - 3} dx \\ &= 2 \ln |x + 1| + 3 \ln |x - 3| + C.\end{aligned}$$

The method for rewriting rational functions as a sum of simpler fractions is called **the method of partial fractions**. In the case of the above example, it consists of finding constants A and B such that

$$\frac{5x - 3}{x^2 - 2x - 3} = \frac{A}{x + 1} + \frac{B}{x - 3}. \quad (1)$$

(Pretend for a moment that we do not know that $A = 2$ and $B = 3$ will work.) We call the fractions $A/(x + 1)$ and $B/(x - 3)$ **partial fractions** because their denominators are only part of the original denominator $x^2 - 2x - 3$. We call A and B **undetermined coefficients** until proper values for them have been found.

To find A and B , we first clear Equation (1) of fractions, obtaining

$$5x - 3 = A(x - 3) + B(x + 1) = (A + B)x - 3A + B.$$

This will be an identity in x if and only if the coefficients of like powers of x on the two sides are equal:

$$A + B = 5, \quad -3A + B = -3.$$

Solving these equations simultaneously gives $A = 2$ and $B = 3$.

General Description of the Method

Success in writing a rational function $f(x)/g(x)$ as a sum of partial fractions depends on two things:

- *The degree of $f(x)$ must be less than the degree of $g(x)$.* That is, the fraction must be proper. If it isn't, divide $f(x)$ by $g(x)$ and work with the remainder term. See Example 3 of this section.
- *We must know the factors of $g(x)$.* In theory, any polynomial with real coefficients can be written as a product of real linear factors and real quadratic factors. In practice, the factors may be hard to find.

Here is how we find the partial fractions of a proper fraction $f(x)/g(x)$ when the factors of g are known.

Method of Partial Fractions ($f(x)/g(x)$ Proper)

1. Let $x - r$ be a linear factor of $g(x)$. Suppose that $(x - r)^m$ is the highest power of $x - r$ that divides $g(x)$. Then, to this factor, assign the sum of the m partial fractions:

$$\frac{A_1}{x - r} + \frac{A_2}{(x - r)^2} + \cdots + \frac{A_m}{(x - r)^m}.$$

Do this for each distinct linear factor of $g(x)$.

2. Let $x^2 + px + q$ be a quadratic factor of $g(x)$. Suppose that $(x^2 + px + q)^n$ is the highest power of this factor that divides $g(x)$. Then, to this factor, assign the sum of the n partial fractions:

$$\frac{B_1x + C_1}{x^2 + px + q} + \frac{B_2x + C_2}{(x^2 + px + q)^2} + \cdots + \frac{B_nx + C_n}{(x^2 + px + q)^n}.$$

Do this for each distinct quadratic factor of $g(x)$ that cannot be factored into linear factors with real coefficients.

3. Set the original fraction $f(x)/g(x)$ equal to the sum of all these partial fractions. Clear the resulting equation of fractions and arrange the terms in decreasing powers of x .
4. Equate the coefficients of corresponding powers of x and solve the resulting equations for the undetermined coefficients.

EXAMPLE 1 Distinct Linear Factors

Evaluate

$$\int \frac{x^2 + 4x + 1}{(x - 1)(x + 1)(x + 3)} dx$$

using partial fractions.

Solution The partial fraction decomposition has the form

$$\frac{x^2 + 4x + 1}{(x - 1)(x + 1)(x + 3)} = \frac{A}{x - 1} + \frac{B}{x + 1} + \frac{C}{x + 3}.$$

To find the values of the undetermined coefficients A , B , and C we clear fractions and get

$$\begin{aligned} x^2 + 4x + 1 &= A(x + 1)(x + 3) + B(x - 1)(x + 3) + C(x - 1)(x + 1) \\ &= (A + B + C)x^2 + (4A + 2B)x + (3A - 3B - C). \end{aligned}$$

The polynomials on both sides of the above equation are identical, so we equate coefficients of like powers of x obtaining

$$\begin{aligned} \text{Coefficient of } x^2: & \quad A + B + C = 1 \\ \text{Coefficient of } x^1: & \quad 4A + 2B = 4 \\ \text{Coefficient of } x^0: & \quad 3A - 3B - C = 1 \end{aligned}$$

There are several ways for solving such a system of linear equations for the unknowns A , B , and C , including elimination of variables, or the use of a calculator or computer. Whatever method is used, the solution is $A = 3/4$, $B = 1/2$, and $C = -1/4$. Hence we have

$$\begin{aligned}\int \frac{x^2 + 4x + 1}{(x-1)(x+1)(x+3)} dx &= \int \left[\frac{3}{4} \frac{1}{x-1} + \frac{1}{2} \frac{1}{x+1} - \frac{1}{4} \frac{1}{x+3} \right] dx \\ &= \frac{3}{4} \ln |x-1| + \frac{1}{2} \ln |x+1| - \frac{1}{4} \ln |x+3| + K,\end{aligned}$$

where K is the arbitrary constant of integration (to avoid confusion with the undetermined coefficient we labeled as C). ■

EXAMPLE 2 A Repeated Linear Factor

Evaluate

$$\int \frac{6x + 7}{(x + 2)^2} dx.$$

Solution First we express the integrand as a sum of partial fractions with undetermined coefficients.

$$\begin{aligned}\frac{6x + 7}{(x + 2)^2} &= \frac{A}{x + 2} + \frac{B}{(x + 2)^2} \\ 6x + 7 &= A(x + 2) + B && \text{Multiply both sides by } (x + 2)^2. \\ &= Ax + (2A + B)\end{aligned}$$

Equating coefficients of corresponding powers of x gives

$$A = 6 \quad \text{and} \quad 2A + B = 12 + B = 7, \quad \text{or} \quad A = 6 \quad \text{and} \quad B = -5.$$

Therefore,

$$\begin{aligned}\int \frac{6x + 7}{(x + 2)^2} dx &= \int \left(\frac{6}{x + 2} - \frac{5}{(x + 2)^2} \right) dx \\ &= 6 \int \frac{dx}{x + 2} - 5 \int (x + 2)^{-2} dx \\ &= 6 \ln |x + 2| + 5(x + 2)^{-1} + C\end{aligned}$$

EXAMPLE 3 Integrating an Improper Fraction

Evaluate

$$\int \frac{2x^3 - 4x^2 - x - 3}{x^2 - 2x - 3} dx.$$

Solution First we divide the denominator into the numerator to get a polynomial plus a proper fraction.

$$\begin{array}{r} 2x \\ x^2 - 2x - 3 \overline{) 2x^3 - 4x^2 - x - 3} \\ \underline{2x^3 - 4x^2 - 6x} \\ 5x - 3 \end{array}$$

Then we write the improper fraction as a polynomial plus a proper fraction.

$$\frac{2x^3 - 4x^2 - x - 3}{x^2 - 2x - 3} = 2x + \frac{5x - 3}{x^2 - 2x - 3}$$

We found the partial fraction decomposition of the fraction on the right in the opening example, so

$$\begin{aligned} \int \frac{2x^3 - 4x^2 - x - 3}{x^2 - 2x - 3} dx &= \int 2x dx + \int \frac{5x - 3}{x^2 - 2x - 3} dx \\ &= \int 2x dx + \int \frac{2}{x + 1} dx + \int \frac{3}{x - 3} dx \\ &= x^2 + 2 \ln |x + 1| + 3 \ln |x - 3| + C. \end{aligned}$$

A quadratic polynomial is **irreducible** if it cannot be written as the product of two linear factors with real coefficients.

EXAMPLE 4 Integrating with an Irreducible Quadratic Factor in the Denominator

Evaluate

$$\int \frac{-2x + 4}{(x^2 + 1)(x - 1)^2} dx$$

using partial fractions.

Solution The denominator has an irreducible quadratic factor as well as a repeated linear factor, so we write

$$\frac{-2x + 4}{(x^2 + 1)(x - 1)^2} = \frac{Ax + B}{x^2 + 1} + \frac{C}{x - 1} + \frac{D}{(x - 1)^2}. \quad (2)$$

Clearing the equation of fractions gives

$$\begin{aligned} -2x + 4 &= (Ax + B)(x - 1)^2 + C(x - 1)(x^2 + 1) + D(x^2 + 1) \\ &= (A + C)x^3 + (-2A + B - C + D)x^2 \\ &\quad + (A - 2B + C)x + (B - C + D). \end{aligned}$$

Equating coefficients of like terms gives

$$\begin{array}{ll} \text{Coefficients of } x^3: & 0 = A + C \\ \text{Coefficients of } x^2: & 0 = -2A + B - C + D \\ \text{Coefficients of } x^1: & -2 = A - 2B + C \\ \text{Coefficients of } x^0: & 4 = B - C + D \end{array}$$

We solve these equations simultaneously to find the values of A , B , C , and D :

$$\begin{aligned} -4 &= -2A, & A &= 2 && \text{Subtract fourth equation from second.} \\ C &= -A = -2 && && \text{From the first equation} \\ B &= 1 && && A = 2 \text{ and } C = -2 \text{ in third equation.} \\ D &= 4 - B + C = 1. && && \text{From the fourth equation} \end{aligned}$$

We substitute these values into Equation (2), obtaining

$$\frac{-2x + 4}{(x^2 + 1)(x - 1)^2} = \frac{2x + 1}{x^2 + 1} - \frac{2}{x - 1} + \frac{1}{(x - 1)^2}.$$

Finally, using the expansion above we can integrate:

$$\begin{aligned} \int \frac{-2x + 4}{(x^2 + 1)(x - 1)^2} dx &= \int \left(\frac{2x + 1}{x^2 + 1} - \frac{2}{x - 1} + \frac{1}{(x - 1)^2} \right) dx \\ &= \int \left(\frac{2x}{x^2 + 1} + \frac{1}{x^2 + 1} - \frac{2}{x - 1} + \frac{1}{(x - 1)^2} \right) dx \\ &= \ln(x^2 + 1) + \tan^{-1} x - 2 \ln|x - 1| - \frac{1}{x - 1} + C. \quad \blacksquare \end{aligned}$$

EXAMPLE 5 A Repeated Irreducible Quadratic Factor

Evaluate

$$\int \frac{dx}{x(x^2 + 1)^2}.$$

Solution The form of the partial fraction decomposition is

$$\frac{1}{x(x^2 + 1)^2} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1} + \frac{Dx + E}{(x^2 + 1)^2}$$

Multiplying by $x(x^2 + 1)^2$, we have

$$\begin{aligned} 1 &= A(x^2 + 1)^2 + (Bx + C)x(x^2 + 1) + (Dx + E)x \\ &= A(x^4 + 2x^2 + 1) + B(x^4 + x^2) + C(x^3 + x) + Dx^2 + Ex \\ &= (A + B)x^4 + Cx^3 + (2A + B + D)x^2 + (C + E)x + A \end{aligned}$$

If we equate coefficients, we get the system

$$A + B = 0, \quad C = 0, \quad 2A + B + D = 0, \quad C + E = 0, \quad A = 1.$$

Solving this system gives $A = 1$, $B = -1$, $C = 0$, $D = -1$, and $E = 0$. Thus,

$$\begin{aligned} \int \frac{dx}{x(x^2 + 1)^2} &= \int \left[\frac{1}{x} + \frac{-x}{x^2 + 1} + \frac{-x}{(x^2 + 1)^2} \right] dx \\ &= \int \frac{dx}{x} - \int \frac{x dx}{x^2 + 1} - \int \frac{x dx}{(x^2 + 1)^2} \\ &= \int \frac{dx}{x} - \frac{1}{2} \int \frac{du}{u} - \frac{1}{2} \int \frac{du}{u^2} \quad \begin{array}{l} u = x^2 + 1, \\ du = 2x dx \end{array} \\ &= \ln|x| - \frac{1}{2} \ln|u| + \frac{1}{2u} + K \\ &= \ln|x| - \frac{1}{2} \ln(x^2 + 1) + \frac{1}{2(x^2 + 1)} + K \\ &= \ln \frac{|x|}{\sqrt{x^2 + 1}} + \frac{1}{2(x^2 + 1)} + K. \quad \blacksquare \end{aligned}$$

HISTORICAL BIOGRAPHY

Oliver Heaviside
(1850–1925)

The Heaviside “Cover-up” Method for Linear Factors

When the degree of the polynomial $f(x)$ is less than the degree of $g(x)$ and

$$g(x) = (x - r_1)(x - r_2) \cdots (x - r_n)$$

is a product of n distinct linear factors, each raised to the first power, there is a quick way to expand $f(x)/g(x)$ by partial fractions.

EXAMPLE 6 Using the Heaviside Method

Find A , B , and C in the partial-fraction expansion

$$\frac{x^2 + 1}{(x - 1)(x - 2)(x - 3)} = \frac{A}{x - 1} + \frac{B}{x - 2} + \frac{C}{x - 3}. \quad (3)$$

Solution If we multiply both sides of Equation (3) by $(x - 1)$ to get

$$\frac{x^2 + 1}{(x - 2)(x - 3)} = A + \frac{B(x - 1)}{x - 2} + \frac{C(x - 1)}{x - 3}$$

and set $x = 1$, the resulting equation gives the value of A :

$$\frac{(1)^2 + 1}{(1 - 2)(1 - 3)} = A + 0 + 0,$$

$$A = 1.$$

Thus, the value of A is the number we would have obtained if we had covered the factor $(x - 1)$ in the denominator of the original fraction

$$\frac{x^2 + 1}{(x - 1)(x - 2)(x - 3)} \quad (4)$$

and evaluated the rest at $x = 1$:

$$A = \frac{(1)^2 + 1}{\boxed{(x - 1)} (1 - 2)(1 - 3)} = \frac{2}{(-1)(-2)} = 1.$$

$\uparrow$
Cover

Similarly, we find the value of B in Equation (3) by covering the factor $(x - 2)$ in Equation (4) and evaluating the rest at $x = 2$:

$$B = \frac{(2)^2 + 1}{(2 - 1) \boxed{(x - 2)} (2 - 3)} = \frac{5}{(1)(-1)} = -5.$$

$\uparrow$
Cover

Finally, C is found by covering the $(x - 3)$ in Equation (4) and evaluating the rest at $x = 3$:

$$C = \frac{(3)^2 + 1}{(3 - 1)(3 - 2) \boxed{(x - 3)}} = \frac{10}{(2)(1)} = 5.$$

$\uparrow$
Cover

Heaviside Method

1. Write the quotient with $g(x)$ factored:

$$\frac{f(x)}{g(x)} = \frac{f(x)}{(x - r_1)(x - r_2) \cdots (x - r_n)}.$$

2. Cover the factors $(x - r_i)$ of $g(x)$ one at a time, each time replacing all the uncovered x 's by the number r_i . This gives a number A_i for each root r_i :

$$\begin{aligned} A_1 &= \frac{f(r_1)}{(r_1 - r_2) \cdots (r_1 - r_n)} \\ A_2 &= \frac{f(r_2)}{(r_2 - r_1)(r_2 - r_3) \cdots (r_2 - r_n)} \\ &\vdots \\ A_n &= \frac{f(r_n)}{(r_n - r_1)(r_n - r_2) \cdots (r_n - r_{n-1})}. \end{aligned}$$

3. Write the partial-fraction expansion of $f(x)/g(x)$ as

$$\frac{f(x)}{g(x)} = \frac{A_1}{(x - r_1)} + \frac{A_2}{(x - r_2)} + \cdots + \frac{A_n}{(x - r_n)}.$$

EXAMPLE 7 Integrating with the Heaviside Method

Evaluate

$$\int \frac{x + 4}{x^3 + 3x^2 - 10x} dx.$$

Solution The degree of $f(x) = x + 4$ is less than the degree of $g(x) = x^3 + 3x^2 - 10x$, and, with $g(x)$ factored,

$$\frac{x + 4}{x^3 + 3x^2 - 10x} = \frac{x + 4}{x(x - 2)(x + 5)}.$$

The roots of $g(x)$ are $r_1 = 0$, $r_2 = 2$, and $r_3 = -5$. We find

$$A_1 = \frac{0 + 4}{\boxed{x} (0 - 2)(0 + 5)} = \frac{4}{(-2)(5)} = -\frac{2}{5}$$

$\uparrow$
 Cover

$$A_2 = \frac{2 + 4}{2 \boxed{(x - 2)} (2 + 5)} = \frac{6}{(2)(7)} = \frac{3}{7}$$

$\uparrow$
 Cover

$$A_3 = \frac{-5 + 4}{(-5)(-5 - 2) \boxed{(x + 5)}} = \frac{-1}{(-5)(-7)} = -\frac{1}{35}.$$

$\uparrow$
 Cover

Therefore,

$$\frac{x+4}{x(x-2)(x+5)} = -\frac{2}{5x} + \frac{3}{7(x-2)} - \frac{1}{35(x+5)},$$

and

$$\int \frac{x+4}{x(x-2)(x+5)} dx = -\frac{2}{5} \ln |x| + \frac{3}{7} \ln |x-2| - \frac{1}{35} \ln |x+5| + C. \quad \blacksquare$$

Other Ways to Determine the Coefficients

Another way to determine the constants that appear in partial fractions is to differentiate, as in the next example. Still another is to assign selected numerical values to x .

EXAMPLE 8 Using Differentiation

Find A , B , and C in the equation

$$\frac{x-1}{(x+1)^3} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3}.$$

Solution We first clear fractions:

$$x-1 = A(x+1)^2 + B(x+1) + C.$$

Substituting $x = -1$ shows $C = -2$. We then differentiate both sides with respect to x , obtaining

$$1 = 2A(x+1) + B.$$

Substituting $x = -1$ shows $B = 1$. We differentiate again to get $0 = 2A$, which shows $A = 0$. Hence,

$$\frac{x-1}{(x+1)^3} = \frac{1}{(x+1)^2} - \frac{2}{(x+1)^3}. \quad \blacksquare$$

In some problems, assigning small values to x such as $x = 0, \pm 1, \pm 2$, to get equations in A , B , and C provides a fast alternative to other methods.

EXAMPLE 9 Assigning Numerical Values to x

Find A , B , and C in

$$\frac{x^2+1}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}.$$

Solution Clear fractions to get

$$x^2+1 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2).$$

Then let $x = 1, 2, 3$ successively to find A , B , and C :

$$\begin{aligned}x = 1: \quad (1)^2 + 1 &= A(-1)(-2) + B(0) + C(0) \\2 &= 2A \\A &= 1\end{aligned}$$

$$\begin{aligned}x = 2: \quad (2)^2 + 1 &= A(0) + B(1)(-1) + C(0) \\5 &= -B \\B &= -5\end{aligned}$$

$$\begin{aligned}x = 3: \quad (3)^2 + 1 &= A(0) + B(0) + C(2)(1) \\10 &= 2C \\C &= 5.\end{aligned}$$

Conclusion:

$$\frac{x^2 + 1}{(x - 1)(x - 2)(x - 3)} = \frac{1}{x - 1} - \frac{5}{x - 2} + \frac{5}{x - 3}.$$



EXERCISES 8.3

Expanding Quotients into Partial Fractions

Expand the quotients in Exercises 1–8 by partial fractions.

1. $\frac{5x - 13}{(x - 3)(x - 2)}$

2. $\frac{5x - 7}{x^2 - 3x + 2}$

3. $\frac{x + 4}{(x + 1)^2}$

4. $\frac{2x + 2}{x^2 - 2x + 1}$

5. $\frac{z + 1}{z^2(z - 1)}$

6. $\frac{z}{z^3 - z^2 - 6z}$

7. $\frac{t^2 + 8}{t^2 - 5t + 6}$

8. $\frac{t^4 + 9}{t^4 + 9t^2}$

Nonrepeated Linear Factors

In Exercises 9–16, express the integrands as a sum of partial fractions and evaluate the integrals.

9. $\int \frac{dx}{1 - x^2}$

10. $\int \frac{dx}{x^2 + 2x}$

11. $\int \frac{x + 4}{x^2 + 5x - 6} dx$

12. $\int \frac{2x + 1}{x^2 - 7x + 12} dx$

13. $\int_4^8 \frac{y dy}{y^2 - 2y - 3}$

14. $\int_{1/2}^1 \frac{y + 4}{y^2 + y} dy$

15. $\int \frac{dt}{t^3 + t^2 - 2t}$

16. $\int \frac{x + 3}{2x^3 - 8x} dx$

Repeated Linear Factors

In Exercises 17–20, express the integrands as a sum of partial fractions and evaluate the integrals.

17. $\int_0^1 \frac{x^3 dx}{x^2 + 2x + 1}$

18. $\int_{-1}^0 \frac{x^3 dx}{x^2 - 2x + 1}$

19. $\int \frac{dx}{(x^2 - 1)^2}$

20. $\int \frac{x^2 dx}{(x - 1)(x^2 + 2x + 1)}$

Irreducible Quadratic Factors

In Exercises 21–28, express the integrands as a sum of partial fractions and evaluate the integrals.

21. $\int_0^1 \frac{dx}{(x + 1)(x^2 + 1)}$

22. $\int_1^{\sqrt{3}} \frac{3t^2 + t + 4}{t^3 + t} dt$

23. $\int \frac{y^2 + 2y + 1}{(y^2 + 1)^2} dy$

24. $\int \frac{8x^2 + 8x + 2}{(4x^2 + 1)^2} dx$

25. $\int \frac{2s + 2}{(s^2 + 1)(s - 1)^3} ds$

26. $\int \frac{s^4 + 81}{s(s^2 + 9)^2} ds$

27. $\int \frac{2\theta^3 + 5\theta^2 + 8\theta + 4}{(\theta^2 + 2\theta + 2)^2} d\theta$

28. $\int \frac{\theta^4 - 4\theta^3 + 2\theta^2 - 3\theta + 1}{(\theta^2 + 1)^3} d\theta$

Improper Fractions

In Exercises 29–34, perform long division on the integrand, write the proper fraction as a sum of partial fractions, and then evaluate the integral.

29. $\int \frac{2x^3 - 2x^2 + 1}{x^2 - x} dx$

30. $\int \frac{x^4}{x^2 - 1} dx$

31. $\int \frac{9x^3 - 3x + 1}{x^3 - x^2} dx$

32. $\int \frac{16x^3}{4x^2 - 4x + 1} dx$

33. $\int \frac{y^4 + y^2 - 1}{y^3 + y} dy$

34. $\int \frac{2y^4}{y^3 - y^2 + y - 1} dy$

Evaluating Integrals

Evaluate the integrals in Exercises 35–40.

35. $\int \frac{e^t dt}{e^{2t} + 3e^t + 2}$ 36. $\int \frac{e^{4t} + 2e^{2t} - e^t}{e^{2t} + 1} dt$
37. $\int \frac{\cos y dy}{\sin^2 y + \sin y - 6}$ 38. $\int \frac{\sin \theta d\theta}{\cos^2 \theta + \cos \theta - 2}$
39. $\int \frac{(x-2)^2 \tan^{-1}(2x) - 12x^3 - 3x}{(4x^2 + 1)(x-2)^2} dx$
40. $\int \frac{(x+1)^2 \tan^{-1}(3x) + 9x^3 + x}{(9x^2 + 1)(x+1)^2} dx$

Initial Value Problems

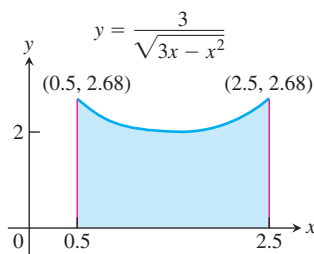
Solve the initial value problems in Exercises 41–44 for x as a function of t .

41. $(t^2 - 3t + 2) \frac{dx}{dt} = 1 \quad (t > 2), \quad x(3) = 0$
42. $(3t^4 + 4t^2 + 1) \frac{dx}{dt} = 2\sqrt{3}, \quad x(1) = -\pi\sqrt{3}/4$
43. $(t^2 + 2t) \frac{dx}{dt} = 2x + 2 \quad (t, x > 0), \quad x(1) = 1$
44. $(t + 1) \frac{dx}{dt} = x^2 + 1 \quad (t > -1), \quad x(0) = \pi/4$

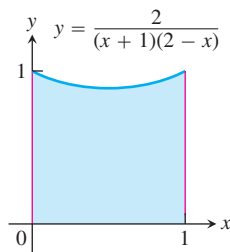
Applications and Examples

In Exercises 45 and 46, find the volume of the solid generated by revolving the shaded region about the indicated axis.

45. The
- x
- axis

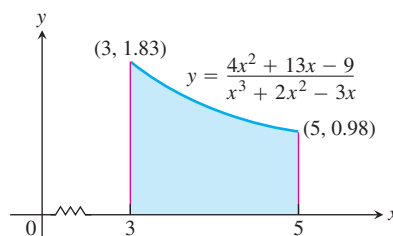


46. The
- y
- axis



- T** 47. Find, to two decimal places, the x -coordinate of the centroid of the region in the first quadrant bounded by the x -axis, the curve $y = \tan^{-1} x$, and the line $x = \sqrt{3}$.

- T** 48. Find the x -coordinate of the centroid of this region to two decimal places.



- T** 49. **Social diffusion** Sociologists sometimes use the phrase “social diffusion” to describe the way information spreads through a population. The information might be a rumor, a cultural fad, or news about a technical innovation. In a sufficiently large population, the number of people x who have the information is treated as a differentiable function of time t , and the rate of diffusion, dx/dt , is assumed to be proportional to the number of people who have the information times the number of people who do not. This leads to the equation

$$\frac{dx}{dt} = kx(N - x),$$

where N is the number of people in the population.Suppose t is in days, $k = 1/250$, and two people start a rumor at time $t = 0$ in a population of $N = 1000$ people.

- a. Find x as a function of t .
- b. When will half the population have heard the rumor? (This is when the rumor will be spreading the fastest.)

- T** 50. **Second-order chemical reactions** Many chemical reactions are the result of the interaction of two molecules that undergo a change to produce a new product. The rate of the reaction typically depends on the concentrations of the two kinds of molecules. If a is the amount of substance A and b is the amount of substance B at time $t = 0$, and if x is the amount of product at time t , then the rate of formation of x may be given by the differential equation

$$\frac{dx}{dt} = k(a - x)(b - x),$$

or

$$\frac{1}{(a - x)(b - x)} \frac{dx}{dt} = k,$$

where k is a constant for the reaction. Integrate both sides of this equation to obtain a relation between x and t (a) if $a = b$, and (b) if $a \neq b$. Assume in each case that $x = 0$ when $t = 0$.

51. **An integral connecting π to the approximation $22/7$**

- a. Evaluate $\int_0^1 \frac{x^4(x-1)^4}{x^2+1} dx$.
- b. How good is the approximation $\pi \approx 22/7$? Find out by expressing $\left(\frac{22}{7} - \pi\right)$ as a percentage of π .

- c. Graph the function $y = \frac{x^4(x-1)^4}{x^2+1}$ for $0 \leq x \leq 1$. Experiment with the range on the y -axis set between 0 and 1, then between 0 and 0.5, and then decreasing the range until the graph can be seen. What do you conclude about the area under the curve?

52. Find the second-degree polynomial $P(x)$ such that $P(0) = 1$, $P'(0) = 0$, and

$$\int \frac{P(x)}{x^3(x-1)^2} dx$$

is a rational function.

8.4

Trigonometric Integrals

Trigonometric integrals involve algebraic combinations of the six basic trigonometric functions. In principle, we can always express such integrals in terms of sines and cosines, but it is often simpler to work with other functions, as in the integral

$$\int \sec^2 x \, dx = \tan x + C.$$

The general idea is to use identities to transform the integrals we have to find into integrals that are easier to work with.

Products of Powers of Sines and Cosines

We begin with integrals of the form:

$$\int \sin^m x \cos^n x \, dx,$$

where m and n are nonnegative integers (positive or zero). We can divide the work into three cases.

Case 1 If m is odd, we write m as $2k + 1$ and use the identity $\sin^2 x = 1 - \cos^2 x$ to obtain

$$\sin^m x = \sin^{2k+1} x = (\sin^2 x)^k \sin x = (1 - \cos^2 x)^k \sin x. \quad (1)$$

Then we combine the single $\sin x$ with dx in the integral and set $\sin x \, dx$ equal to $-d(\cos x)$.

Case 2 If m is even and n is odd in $\int \sin^m x \cos^n x \, dx$, we write n as $2k + 1$ and use the identity $\cos^2 x = 1 - \sin^2 x$ to obtain

$$\cos^n x = \cos^{2k+1} x = (\cos^2 x)^k \cos x = (1 - \sin^2 x)^k \cos x.$$

We then combine the single $\cos x$ with dx and set $\cos x \, dx$ equal to $d(\sin x)$.

Case 3 If both m and n are even in $\int \sin^m x \cos^n x \, dx$, we substitute

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2} \quad (2)$$

to reduce the integrand to one in lower powers of $\cos 2x$.

Here are some examples illustrating each case.

EXAMPLE 1 m is Odd

Evaluate

$$\int \sin^3 x \cos^2 x \, dx.$$

Solution

$$\begin{aligned}
 \int \sin^3 x \cos^2 x \, dx &= \int \sin^2 x \cos^2 x \sin x \, dx \\
 &= \int (1 - \cos^2 x) \cos^2 x (-d(\cos x)) \\
 &= \int (1 - u^2)(u^2)(-du) && u = \cos x \\
 &= \int (u^4 - u^2) \, du \\
 &= \frac{u^5}{5} - \frac{u^3}{3} + C \\
 &= \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C.
 \end{aligned}$$

EXAMPLE 2 m is Even and n is Odd

Evaluate

$$\int \cos^5 x \, dx.$$

Solution

$$\begin{aligned}
 \int \cos^5 x \, dx &= \int \cos^4 x \cos x \, dx = \int (1 - \sin^2 x)^2 d(\sin x) && m = 0 \\
 &= \int (1 - u^2)^2 \, du && u = \sin x \\
 &= \int (1 - 2u^2 + u^4) \, du \\
 &= u - \frac{2}{3}u^3 + \frac{1}{5}u^5 + C = \sin x - \frac{2}{3}\sin^3 x + \frac{1}{5}\sin^5 x + C.
 \end{aligned}$$

EXAMPLE 3 m and n are Both Even

Evaluate

$$\int \sin^2 x \cos^4 x \, dx.$$

Solution

$$\begin{aligned}
 \int \sin^2 x \cos^4 x \, dx &= \int \left(\frac{1 - \cos 2x}{2} \right) \left(\frac{1 + \cos 2x}{2} \right)^2 dx \\
 &= \frac{1}{8} \int (1 - \cos 2x)(1 + 2\cos 2x + \cos^2 2x) \, dx \\
 &= \frac{1}{8} \int (1 + \cos 2x - \cos^2 2x - \cos^3 2x) \, dx \\
 &= \frac{1}{8} \left[x + \frac{1}{2} \sin 2x - \int (\cos^2 2x + \cos^3 2x) \, dx \right].
 \end{aligned}$$

For the term involving $\cos^2 2x$ we use

$$\begin{aligned}\int \cos^2 2x \, dx &= \frac{1}{2} \int (1 + \cos 4x) \, dx \\ &= \frac{1}{2} \left(x + \frac{1}{4} \sin 4x \right).\end{aligned}$$

Omitting the constant of integration until the final result

For the $\cos^3 2x$ term we have

$$\begin{aligned}\int \cos^3 2x \, dx &= \int (1 - \sin^2 2x) \cos 2x \, dx \\ &= \frac{1}{2} \int (1 - u^2) \, du = \frac{1}{2} \left(\sin 2x - \frac{1}{3} \sin^3 2x \right).\end{aligned}$$

$u = \sin 2x,$
 $du = 2 \cos 2x \, dx$
Again omitting C

Combining everything and simplifying we get

$$\int \sin^2 x \cos^4 x \, dx = \frac{1}{16} \left(x - \frac{1}{4} \sin 4x + \frac{1}{3} \sin^3 2x \right) + C. \quad \blacksquare$$

Eliminating Square Roots

In the next example, we use the identity $\cos^2 \theta = (1 + \cos 2\theta)/2$ to eliminate a square root.

EXAMPLE 4 Evaluate

$$\int_0^{\pi/4} \sqrt{1 + \cos 4x} \, dx.$$

Solution To eliminate the square root we use the identity

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}, \quad \text{or} \quad 1 + \cos 2\theta = 2 \cos^2 \theta.$$

With $\theta = 2x$, this becomes

$$1 + \cos 4x = 2 \cos^2 2x.$$

Therefore,

$$\begin{aligned}\int_0^{\pi/4} \sqrt{1 + \cos 4x} \, dx &= \int_0^{\pi/4} \sqrt{2 \cos^2 2x} \, dx = \int_0^{\pi/4} \sqrt{2} \sqrt{\cos^2 2x} \, dx \\ &= \sqrt{2} \int_0^{\pi/4} |\cos 2x| \, dx = \sqrt{2} \int_0^{\pi/4} \cos 2x \, dx \\ &= \sqrt{2} \left[\frac{\sin 2x}{2} \right]_0^{\pi/4} = \frac{\sqrt{2}}{2} [1 - 0] = \frac{\sqrt{2}}{2}.\end{aligned}$$

$\cos 2x \geq 0$
on $[0, \pi/4]$

Integrals of Powers of $\tan x$ and $\sec x$

We know how to integrate the tangent and secant and their squares. To integrate higher powers we use the identities $\tan^2 x = \sec^2 x - 1$ and $\sec^2 x = \tan^2 x + 1$, and integrate by parts when necessary to reduce the higher powers to lower powers.

EXAMPLE 5 Evaluate

$$\int \tan^4 x \, dx.$$

Solution

$$\begin{aligned} \int \tan^4 x \, dx &= \int \tan^2 x \cdot \tan^2 x \, dx = \int \tan^2 x \cdot (\sec^2 x - 1) \, dx \\ &= \int \tan^2 x \sec^2 x \, dx - \int \tan^2 x \, dx \\ &= \int \tan^2 x \sec^2 x \, dx - \int (\sec^2 x - 1) \, dx \\ &= \int \tan^2 x \sec^2 x \, dx - \int \sec^2 x \, dx + \int dx. \end{aligned}$$

In the first integral, we let

$$u = \tan x, \quad du = \sec^2 x \, dx$$

and have

$$\int u^2 \, du = \frac{1}{3} u^3 + C_1.$$

The remaining integrals are standard forms, so

$$\int \tan^4 x \, dx = \frac{1}{3} \tan^3 x - \tan x + x + C. \quad \blacksquare$$

EXAMPLE 6 Evaluate

$$\int \sec^3 x \, dx.$$

Solution We integrate by parts, using

$$u = \sec x, \quad dv = \sec^2 x \, dx, \quad v = \tan x, \quad du = \sec x \tan x \, dx.$$

Then

$$\begin{aligned} \int \sec^3 x \, dx &= \sec x \tan x - \int (\tan x)(\sec x \tan x \, dx) \\ &= \sec x \tan x - \int (\sec^2 x - 1) \sec x \, dx && \tan^2 x = \sec^2 x - 1 \\ &= \sec x \tan x + \int \sec x \, dx - \int \sec^3 x \, dx. \end{aligned}$$

Combining the two secant-cubed integrals gives

$$2 \int \sec^3 x \, dx = \sec x \tan x + \int \sec x \, dx$$

and

$$\int \sec^3 x \, dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C. \quad \blacksquare$$

Products of Sines and Cosines

The integrals

$$\int \sin mx \sin nx \, dx, \quad \int \sin mx \cos nx \, dx, \quad \text{and} \quad \int \cos mx \cos nx \, dx$$

arise in many places where trigonometric functions are applied to problems in mathematics and science. We can evaluate these integrals through integration by parts, but two such integrations are required in each case. It is simpler to use the identities

$$\sin mx \sin nx = \frac{1}{2} [\cos (m - n)x - \cos (m + n)x], \quad (3)$$

$$\sin mx \cos nx = \frac{1}{2} [\sin (m - n)x + \sin (m + n)x], \quad (4)$$

$$\cos mx \cos nx = \frac{1}{2} [\cos (m - n)x + \cos (m + n)x]. \quad (5)$$

These come from the angle sum formulas for the sine and cosine functions (Section 1.6). They give functions whose antiderivatives are easily found.

EXAMPLE 7 Evaluate

$$\int \sin 3x \cos 5x \, dx.$$

Solution From Equation (4) with $m = 3$ and $n = 5$ we get

$$\begin{aligned} \int \sin 3x \cos 5x \, dx &= \frac{1}{2} \int [\sin (-2x) + \sin 8x] \, dx \\ &= \frac{1}{2} \int (\sin 8x - \sin 2x) \, dx \\ &= -\frac{\cos 8x}{16} + \frac{\cos 2x}{4} + C. \quad \blacksquare \end{aligned}$$

EXERCISES 8.4

Products of Powers of Sines and Cosines

Evaluate the integrals in Exercises 1–14.

1. $\int_0^{\pi/2} \sin^5 x \, dx$

2. $\int_0^{\pi} \sin^5 \frac{x}{2} \, dx$

3. $\int_{-\pi/2}^{\pi/2} \cos^3 x \, dx$

5. $\int_0^{\pi/2} \sin^7 y \, dy$

4. $\int_0^{\pi/6} 3 \cos^5 3x \, dx$

6. $\int_0^{\pi/2} 7 \cos^7 t \, dt$

7. $\int_0^{\pi} 8 \sin^4 x \, dx$ 8. $\int_0^1 8 \cos^4 2\pi x \, dx$
9. $\int_{-\pi/4}^{\pi/4} 16 \sin^2 x \cos^2 x \, dx$ 10. $\int_0^{\pi} 8 \sin^4 y \cos^2 y \, dy$
11. $\int_0^{\pi/2} 35 \sin^4 x \cos^3 x \, dx$ 12. $\int_0^{\pi} \sin 2x \cos^2 2x \, dx$
13. $\int_0^{\pi/4} 8 \cos^3 2\theta \sin 2\theta \, d\theta$ 14. $\int_0^{\pi/2} \sin^2 2\theta \cos^3 2\theta \, d\theta$

Integrals with Square Roots

Evaluate the integrals in Exercises 15–22.

15. $\int_0^{2\pi} \sqrt{\frac{1 - \cos x}{2}} \, dx$ 16. $\int_0^{\pi} \sqrt{1 - \cos 2x} \, dx$
17. $\int_0^{\pi} \sqrt{1 - \sin^2 t} \, dt$ 18. $\int_0^{\pi} \sqrt{1 - \cos^2 \theta} \, d\theta$
19. $\int_{-\pi/4}^{\pi/4} \sqrt{1 + \tan^2 x} \, dx$ 20. $\int_{-\pi/4}^{\pi/4} \sqrt{\sec^2 x - 1} \, dx$
21. $\int_0^{\pi/2} \theta \sqrt{1 - \cos 2\theta} \, d\theta$ 22. $\int_{-\pi}^{\pi} (1 - \cos^2 t)^{3/2} \, dt$

Powers of Tan x and Sec x

Evaluate the integrals in Exercises 23–32.

23. $\int_{-\pi/3}^0 2 \sec^3 x \, dx$ 24. $\int e^x \sec^3 e^x \, dx$
25. $\int_0^{\pi/4} \sec^4 \theta \, d\theta$ 26. $\int_0^{\pi/12} 3 \sec^4 3x \, dx$
27. $\int_{\pi/4}^{\pi/2} \csc^4 \theta \, d\theta$ 28. $\int_{\pi/2}^{\pi} 3 \csc^4 \frac{\theta}{2} \, d\theta$
29. $\int_0^{\pi/4} 4 \tan^3 x \, dx$ 30. $\int_{-\pi/4}^{\pi/4} 6 \tan^4 x \, dx$
31. $\int_{\pi/6}^{\pi/3} \cot^3 x \, dx$ 32. $\int_{\pi/4}^{\pi/2} 8 \cot^4 t \, dt$

Products of Sines and Cosines

Evaluate the integrals in Exercises 33–38.

33. $\int_{-\pi}^0 \sin 3x \cos 2x \, dx$ 34. $\int_0^{\pi/2} \sin 2x \cos 3x \, dx$

35. $\int_{-\pi}^{\pi} \sin 3x \sin 3x \, dx$ 36. $\int_0^{\pi/2} \sin x \cos x \, dx$
37. $\int_0^{\pi} \cos 3x \cos 4x \, dx$ 38. $\int_{-\pi/2}^{\pi/2} \cos x \cos 7x \, dx$

Theory and Examples

39. **Surface area** Find the area of the surface generated by revolving the arc

$$x = t^{2/3}, \quad y = t^2/2, \quad 0 \leq t \leq 2,$$

about the x -axis.

40. **Arc length** Find the length of the curve

$$y = \ln(\cos x), \quad 0 \leq x \leq \pi/3.$$

41. **Arc length** Find the length of the curve

$$y = \ln(\sec x), \quad 0 \leq x \leq \pi/4.$$

42. **Center of gravity** Find the center of gravity of the region bounded by the x -axis, the curve $y = \sec x$, and the lines $x = -\pi/4$, $x = \pi/4$.

43. **Volume** Find the volume generated by revolving one arch of the curve $y = \sin x$ about the x -axis.

44. **Area** Find the area between the x -axis and the curve $y = \sqrt{1 + \cos 4x}$, $0 \leq x \leq \pi$.

45. **Orthogonal functions** Two functions f and g are said to be **orthogonal** on an interval $a \leq x \leq b$ if $\int_a^b f(x)g(x) \, dx = 0$.

a. Prove that $\sin mx$ and $\sin nx$ are orthogonal on any interval of length 2π provided m and n are integers such that $m^2 \neq n^2$.

b. Prove the same for $\cos mx$ and $\cos nx$.

c. Prove the same for $\sin mx$ and $\cos nx$ even if $m = n$.

46. **Fourier series** A finite Fourier series is given by the sum

$$\begin{aligned} f(x) &= \sum_{n=1}^N a_n \sin nx \\ &= a_1 \sin x + a_2 \sin 2x + \cdots + a_N \sin Nx \end{aligned}$$

Show that the m th coefficient a_m is given by the formula

$$a_m = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin mx \, dx.$$

8.5

Trigonometric Substitutions

Trigonometric substitutions can be effective in transforming integrals involving $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, and $\sqrt{x^2 - a^2}$ into integrals we can evaluate directly.

Three Basic Substitutions

The most common substitutions are $x = a \tan \theta$, $x = a \sin \theta$, and $x = a \sec \theta$. They come from the reference right triangles in Figure 8.2.

With $x = a \tan \theta$,

$$a^2 + x^2 = a^2 + a^2 \tan^2 \theta = a^2(1 + \tan^2 \theta) = a^2 \sec^2 \theta.$$

With $x = a \sin \theta$,

$$a^2 - x^2 = a^2 - a^2 \sin^2 \theta = a^2(1 - \sin^2 \theta) = a^2 \cos^2 \theta.$$

With $x = a \sec \theta$,

$$x^2 - a^2 = a^2 \sec^2 \theta - a^2 = a^2(\sec^2 \theta - 1) = a^2 \tan^2 \theta.$$

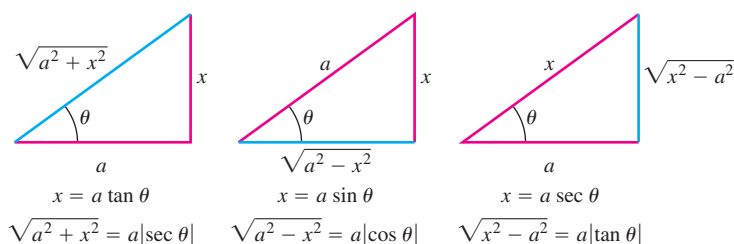


FIGURE 8.2 Reference triangles for the three basic substitutions identifying the sides labeled x and a for each substitution.

We want any substitution we use in an integration to be reversible so that we can change back to the original variable afterward. For example, if $x = a \tan \theta$, we want to be able to set $\theta = \tan^{-1}(x/a)$ after the integration takes place. If $x = a \sin \theta$, we want to be able to set $\theta = \sin^{-1}(x/a)$ when we're done, and similarly for $x = a \sec \theta$.

As we know from Section 7.7, the functions in these substitutions have inverses only for selected values of θ (Figure 8.3). For reversibility,

$$x = a \tan \theta \quad \text{requires} \quad \theta = \tan^{-1}\left(\frac{x}{a}\right) \quad \text{with} \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2},$$

$$x = a \sin \theta \quad \text{requires} \quad \theta = \sin^{-1}\left(\frac{x}{a}\right) \quad \text{with} \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2},$$

$$x = a \sec \theta \quad \text{requires} \quad \theta = \sec^{-1}\left(\frac{x}{a}\right) \quad \text{with} \quad \begin{cases} 0 \leq \theta < \frac{\pi}{2} & \text{if } \frac{x}{a} \geq 1, \\ \frac{\pi}{2} < \theta \leq \pi & \text{if } \frac{x}{a} \leq -1. \end{cases}$$

To simplify calculations with the substitution $x = a \sec \theta$, we will restrict its use to integrals in which $x/a \geq 1$. This will place θ in $[0, \pi/2)$ and make $\tan \theta \geq 0$. We will then have $\sqrt{x^2 - a^2} = \sqrt{a^2 \tan^2 \theta} = |a \tan \theta| = a \tan \theta$, free of absolute values, provided $a > 0$.

EXAMPLE 1 Using the Substitution $x = a \tan \theta$

Evaluate

$$\int \frac{dx}{\sqrt{4 + x^2}}.$$

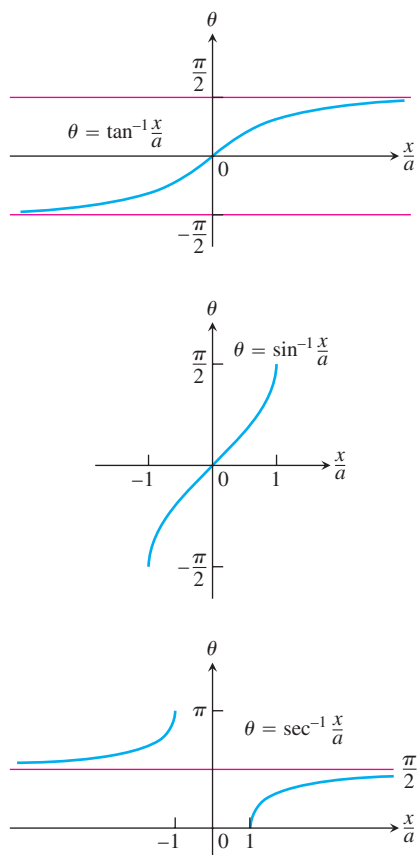


FIGURE 8.3 The arctangent, arcsine, and arcsecant of x/a , graphed as functions of x/a .

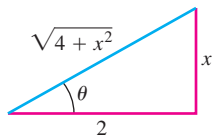


FIGURE 8.4 Reference triangle for $x = 2 \tan \theta$ (Example 1):

$$\tan \theta = \frac{x}{2}$$

and

$$\sec \theta = \frac{\sqrt{4 + x^2}}{2}.$$

Solution We set

$$x = 2 \tan \theta, \quad dx = 2 \sec^2 \theta d\theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2},$$

$$4 + x^2 = 4 + 4 \tan^2 \theta = 4(1 + \tan^2 \theta) = 4 \sec^2 \theta.$$

Then

$$\begin{aligned} \int \frac{dx}{\sqrt{4 + x^2}} &= \int \frac{2 \sec^2 \theta d\theta}{\sqrt{4 \sec^2 \theta}} = \int \frac{\sec^2 \theta d\theta}{|\sec \theta|} && \sqrt{\sec^2 \theta} = |\sec \theta| \\ &= \int \sec \theta d\theta && \sec \theta > 0 \text{ for } -\frac{\pi}{2} < \theta < \frac{\pi}{2} \\ &= \ln |\sec \theta + \tan \theta| + C \\ &= \ln \left| \frac{\sqrt{4 + x^2}}{2} + \frac{x}{2} \right| + C && \text{From Fig. 8.4} \\ &= \ln |\sqrt{4 + x^2} + x| + C'. && \text{Taking } C' = C - \ln 2 \end{aligned}$$

Notice how we expressed $\ln |\sec \theta + \tan \theta|$ in terms of x : We drew a reference triangle for the original substitution $x = 2 \tan \theta$ (Figure 8.4) and read the ratios from the triangle. ■

EXAMPLE 2 Using the Substitution $x = a \sin \theta$

Evaluate

$$\int \frac{x^2 dx}{\sqrt{9 - x^2}}.$$

Solution We set

$$x = 3 \sin \theta, \quad dx = 3 \cos \theta d\theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

$$9 - x^2 = 9 - 9 \sin^2 \theta = 9(1 - \sin^2 \theta) = 9 \cos^2 \theta.$$

Then

$$\begin{aligned} \int \frac{x^2 dx}{\sqrt{9 - x^2}} &= \int \frac{9 \sin^2 \theta \cdot 3 \cos \theta d\theta}{|3 \cos \theta|} \\ &= 9 \int \sin^2 \theta d\theta && \cos \theta > 0 \text{ for } -\frac{\pi}{2} < \theta < \frac{\pi}{2} \\ &= 9 \int \frac{1 - \cos 2\theta}{2} d\theta \\ &= \frac{9}{2} \left(\theta - \frac{\sin 2\theta}{2} \right) + C \\ &= \frac{9}{2} (\theta - \sin \theta \cos \theta) + C && \sin 2\theta = 2 \sin \theta \cos \theta \\ &= \frac{9}{2} \left(\sin^{-1} \frac{x}{3} - \frac{x}{3} \cdot \frac{\sqrt{9 - x^2}}{3} \right) + C && \text{Fig. 8.5} \\ &= \frac{9}{2} \sin^{-1} \frac{x}{3} - \frac{x}{2} \sqrt{9 - x^2} + C. \end{aligned}$$

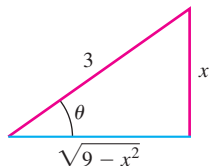


FIGURE 8.5 Reference triangle for $x = 3 \sin \theta$ (Example 2):

$$\sin \theta = \frac{x}{3}$$

and

$$\cos \theta = \frac{\sqrt{9 - x^2}}{3}.$$

EXAMPLE 3 Using the Substitution $x = a \sec \theta$

Evaluate

$$\int \frac{dx}{\sqrt{25x^2 - 4}}, \quad x > \frac{2}{5}.$$

Solution We first rewrite the radical as

$$\begin{aligned}\sqrt{25x^2 - 4} &= \sqrt{25\left(x^2 - \frac{4}{25}\right)} \\ &= 5\sqrt{x^2 - \left(\frac{2}{5}\right)^2}\end{aligned}$$

to put the radicand in the form $x^2 - a^2$. We then substitute

$$x = \frac{2}{5} \sec \theta, \quad dx = \frac{2}{5} \sec \theta \tan \theta \, d\theta, \quad 0 < \theta < \frac{\pi}{2}$$

$$\begin{aligned}x^2 - \left(\frac{2}{5}\right)^2 &= \frac{4}{25} \sec^2 \theta - \frac{4}{25} \\ &= \frac{4}{25} (\sec^2 \theta - 1) = \frac{4}{25} \tan^2 \theta\end{aligned}$$

$$\sqrt{x^2 - \left(\frac{2}{5}\right)^2} = \frac{2}{5} |\tan \theta| = \frac{2}{5} \tan \theta. \quad \begin{array}{l} \tan \theta > 0 \text{ for} \\ 0 < \theta < \pi/2 \end{array}$$

With these substitutions, we have

$$\begin{aligned}\int \frac{dx}{\sqrt{25x^2 - 4}} &= \int \frac{dx}{5\sqrt{x^2 - (4/25)}} = \int \frac{(2/5) \sec \theta \tan \theta \, d\theta}{5 \cdot (2/5) \tan \theta} \\ &= \frac{1}{5} \int \sec \theta \, d\theta = \frac{1}{5} \ln |\sec \theta + \tan \theta| + C \\ &= \frac{1}{5} \ln \left| \frac{5x}{2} + \frac{\sqrt{25x^2 - 4}}{2} \right| + C.\end{aligned}$$

Fig. 8.6

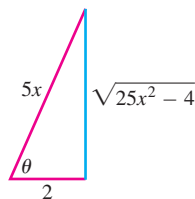


FIGURE 8.6 If $x = (2/5)\sec \theta$, $0 < \theta < \pi/2$, then $\theta = \sec^{-1}(5x/2)$, and we can read the values of the other trigonometric functions of θ from this right triangle (Example 3).

A trigonometric substitution can sometimes help us to evaluate an integral containing an integer power of a quadratic binomial, as in the next example.

EXAMPLE 4 Finding the Volume of a Solid of Revolution

Find the volume of the solid generated by revolving about the x -axis the region bounded by the curve $y = 4/(x^2 + 4)$, the x -axis, and the lines $x = 0$ and $x = 2$.

Solution We sketch the region (Figure 8.7) and use the disk method:

$$V = \int_0^2 \pi [R(x)]^2 \, dx = 16\pi \int_0^2 \frac{dx}{(x^2 + 4)^2}. \quad R(x) = \frac{4}{x^2 + 4}$$

To evaluate the integral, we set

$$\begin{aligned}x &= 2 \tan \theta, & dx &= 2 \sec^2 \theta \, d\theta, & \theta &= \tan^{-1} \frac{x}{2}, \\ x^2 + 4 &= 4 \tan^2 \theta + 4 = 4(\tan^2 \theta + 1) = 4 \sec^2 \theta\end{aligned}$$

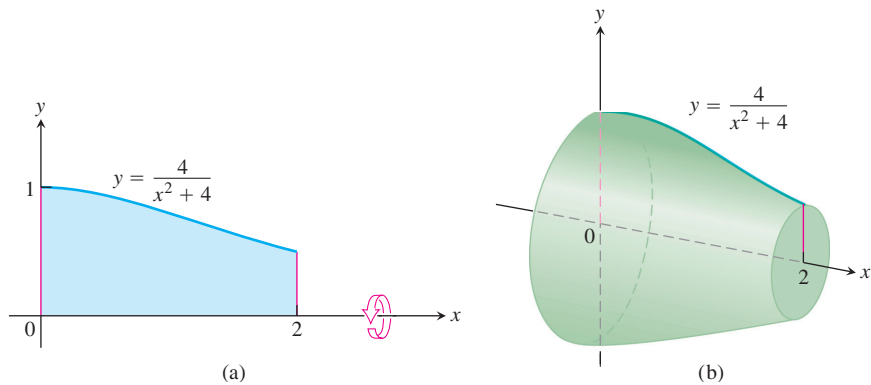
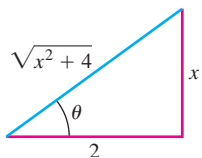


FIGURE 8.7 The region (a) and solid (b) in Example 4.


 FIGURE 8.8 Reference triangle for $x = 2 \tan \theta$ (Example 4).

(Figure 8.8). With these substitutions,

$$\begin{aligned}
 V &= 16\pi \int_0^2 \frac{dx}{(x^2 + 4)^2} \\
 &= 16\pi \int_0^{\pi/4} \frac{2 \sec^2 \theta d\theta}{(4 \sec^2 \theta)^2} && \theta = 0 \text{ when } x = 0; \\
 &= 16\pi \int_0^{\pi/4} \frac{2 \sec^2 \theta d\theta}{16 \sec^4 \theta} = \pi \int_0^{\pi/4} 2 \cos^2 \theta d\theta && \theta = \pi/4 \text{ when } x = 2 \\
 &= \pi \int_0^{\pi/4} (1 + \cos 2\theta) d\theta = \pi \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\pi/4} && 2 \cos^2 \theta = 1 + \cos 2\theta \\
 &= \pi \left[\frac{\pi}{4} + \frac{1}{2} \right] \approx 4.04.
 \end{aligned}$$

EXAMPLE 5 Finding the Area of an Ellipse

Find the area enclosed by the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Solution Because the ellipse is symmetric with respect to both axes, the total area A is four times the area in the first quadrant (Figure 8.9). Solving the equation of the ellipse for $y \geq 0$, we get

$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2} = \frac{a^2 - x^2}{a^2},$$

or

$$y = \frac{b}{a} \sqrt{a^2 - x^2} \quad 0 \leq x \leq a$$

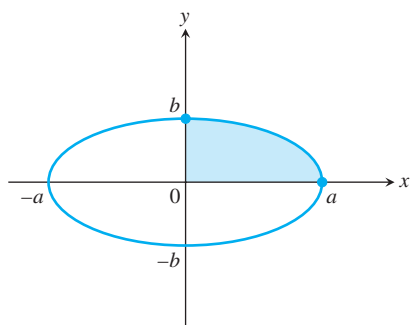


FIGURE 8.9 The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in Example 5.

The area of the ellipse is

$$\begin{aligned} A &= 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx \\ &= 4 \frac{b}{a} \int_0^{\pi/2} a \cos \theta \cdot a \cos \theta d\theta && \begin{array}{l} x = a \sin \theta, dx = a \cos \theta d\theta, \\ \theta = 0 \text{ when } x = 0; \\ \theta = \pi/2 \text{ when } x = a \end{array} \\ &= 4ab \int_0^{\pi/2} \cos^2 \theta d\theta \\ &= 4ab \int_0^{\pi/2} \frac{1 + \cos 2\theta}{2} d\theta \\ &= 2ab \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\pi/2} \\ &= 2ab \left[\frac{\pi}{2} + 0 - 0 \right] = \pi ab. \end{aligned}$$

If $a = b = r$ we get that the area of a circle with radius r is πr^2 . ■

EXERCISES 8.5

Basic Trigonometric Substitutions

Evaluate the integrals in Exercises 1–28.

1. $\int \frac{dy}{\sqrt{9+y^2}}$
2. $\int \frac{3 dy}{\sqrt{1+9y^2}}$
3. $\int_{-2}^2 \frac{dx}{4+x^2}$
4. $\int_0^2 \frac{dx}{8+2x^2}$
5. $\int_0^{3/2} \frac{dx}{\sqrt{9-x^2}}$
6. $\int_0^{1/2\sqrt{2}} \frac{2 dx}{\sqrt{1-4x^2}}$
7. $\int \sqrt{25-t^2} dt$
8. $\int \sqrt{1-9t^2} dt$
9. $\int \frac{dx}{\sqrt{4x^2-49}}, \quad x > \frac{7}{2}$
10. $\int \frac{5 dx}{\sqrt{25x^2-9}}, \quad x > \frac{3}{5}$
11. $\int \frac{\sqrt{y^2-49}}{y} dy, \quad y > 7$
12. $\int \frac{\sqrt{y^2-25}}{y^3} dy, \quad y > 5$
13. $\int \frac{dx}{x^2\sqrt{x^2-1}}, \quad x > 1$
14. $\int \frac{2 dx}{x^3\sqrt{x^2-1}}, \quad x > 1$
15. $\int \frac{x^3 dx}{\sqrt{x^2+4}}$
16. $\int \frac{dx}{x^2\sqrt{x^2+1}}$
17. $\int \frac{8 dw}{w^2\sqrt{4-w^2}}$
18. $\int \frac{\sqrt{9-w^2}}{w^2} dw$
19. $\int_0^{\sqrt{3}/2} \frac{4x^2 dx}{(1-x^2)^{3/2}}$
20. $\int_0^1 \frac{dx}{(4-x^2)^{3/2}}$
21. $\int \frac{dx}{(x^2-1)^{3/2}}, \quad x > 1$
22. $\int \frac{x^2 dx}{(x^2-1)^{5/2}}, \quad x > 1$

23. $\int \frac{(1-x^2)^{3/2}}{x^6} dx$
24. $\int \frac{(1-x^2)^{1/2}}{x^4} dx$
25. $\int \frac{8 dx}{(4x^2+1)^2}$
26. $\int \frac{6 dt}{(9t^2+1)^2}$
27. $\int \frac{v^2 dv}{(1-v^2)^{5/2}}$
28. $\int \frac{(1-r^2)^{5/2}}{r^8} dr$

In Exercises 29–36, use an appropriate substitution and then a trigonometric substitution to evaluate the integrals.

29. $\int_0^{\ln 4} \frac{e^t dt}{\sqrt{e^{2t}+9}}$
30. $\int_{\ln(3/4)}^{\ln(4/3)} \frac{e^t dt}{(1+e^{2t})^{3/2}}$
31. $\int_{1/12}^{1/4} \frac{2 dt}{\sqrt{t}+4t\sqrt{t}}$
32. $\int_1^e \frac{dy}{y\sqrt{1+(\ln y)^2}}$
33. $\int \frac{dx}{x\sqrt{x^2-1}}$
34. $\int \frac{dx}{1+x^2}$
35. $\int \frac{x dx}{\sqrt{x^2-1}}$
36. $\int \frac{dx}{\sqrt{1-x^2}}$

Initial Value Problems

Solve the initial value problems in Exercises 37–40 for y as a function of x .

37. $x \frac{dy}{dx} = \sqrt{x^2-4}, \quad x \geq 2, \quad y(2) = 0$
38. $\sqrt{x^2-9} \frac{dy}{dx} = 1, \quad x > 3, \quad y(5) = \ln 3$
39. $(x^2+4) \frac{dy}{dx} = 3, \quad y(2) = 0$
40. $(x^2+1)^2 \frac{dy}{dx} = \sqrt{x^2+1}, \quad y(0) = 1$

Applications

41. Find the area of the region in the first quadrant that is enclosed by the coordinate axes and the curve $y = \sqrt{9 - x^2}/3$.
42. Find the volume of the solid generated by revolving about the x -axis the region in the first quadrant enclosed by the coordinate axes, the curve $y = 2/(1 + x^2)$, and the line $x = 1$.

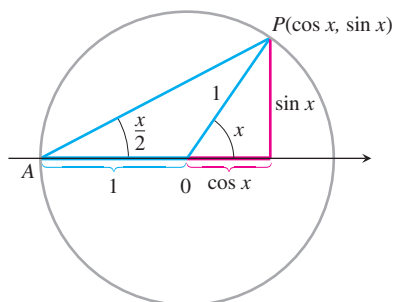
The Substitution $z = \tan(x/2)$

The substitution

$$z = \tan \frac{x}{2} \quad (1)$$

reduces the problem of integrating a rational expression in $\sin x$ and $\cos x$ to a problem of integrating a rational function of z . This in turn can be integrated by partial fractions.

From the accompanying figure



we can read the relation

$$\tan \frac{x}{2} = \frac{\sin x}{1 + \cos x}.$$

To see the effect of the substitution, we calculate

$$\begin{aligned} \cos x &= 2 \cos^2 \left(\frac{x}{2} \right) - 1 = \frac{2}{\sec^2(x/2)} - 1 \\ &= \frac{2}{1 + \tan^2(x/2)} - 1 = \frac{2}{1 + z^2} - 1 \\ \cos x &= \frac{1 - z^2}{1 + z^2}, \end{aligned} \quad (2)$$

and

$$\begin{aligned} \sin x &= 2 \sin \frac{x}{2} \cos \frac{x}{2} = 2 \frac{\sin(x/2)}{\cos(x/2)} \cdot \cos^2 \left(\frac{x}{2} \right) \\ &= 2 \tan \frac{x}{2} \cdot \frac{1}{\sec^2(x/2)} = \frac{2 \tan(x/2)}{1 + \tan^2(x/2)} \\ \sin x &= \frac{2z}{1 + z^2}. \end{aligned} \quad (3)$$

Finally, $x = 2 \tan^{-1} z$, so

$$dx = \frac{2 dz}{1 + z^2}. \quad (4)$$

Examples

$$\begin{aligned} \text{a. } \int \frac{1}{1 + \cos x} dx &= \int \frac{1 + z^2}{2} \frac{2 dz}{1 + z^2} \\ &= \int dz = z + C \\ &= \tan \left(\frac{x}{2} \right) + C \\ \text{b. } \int \frac{1}{2 + \sin x} dx &= \int \frac{1 + z^2}{2 + 2z + 2z^2} \frac{2 dz}{1 + z^2} \\ &= \int \frac{dz}{z^2 + z + 1} = \int \frac{dz}{(z + (1/2))^2 + 3/4} \\ &= \int \frac{du}{u^2 + a^2} \\ &= \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + C \\ &= \frac{2}{\sqrt{3}} \tan^{-1} \frac{2z + 1}{\sqrt{3}} + C \\ &= \frac{2}{\sqrt{3}} \tan^{-1} \frac{1 + 2 \tan(x/2)}{\sqrt{3}} + C \end{aligned}$$

Use the substitutions in Equations (1)–(4) to evaluate the integrals in Exercises 43–50. Integrals like these arise in calculating the average angular velocity of the output shaft of a universal joint when the input and output shafts are not aligned.

$$\begin{aligned} 43. \int \frac{dx}{1 - \sin x} & \quad 44. \int \frac{dx}{1 + \sin x + \cos x} \\ 45. \int_0^{\pi/2} \frac{dx}{1 + \sin x} & \quad 46. \int_{\pi/3}^{\pi/2} \frac{dx}{1 - \cos x} \\ 47. \int_0^{\pi/2} \frac{d\theta}{2 + \cos \theta} & \quad 48. \int_{\pi/2}^{2\pi/3} \frac{\cos \theta d\theta}{\sin \theta \cos \theta + \sin \theta} \\ 49. \int \frac{dt}{\sin t - \cos t} & \quad 50. \int \frac{\cos t dt}{1 - \cos t} \end{aligned}$$

Use the substitution $z = \tan(\theta/2)$ to evaluate the integrals in Exercises 51 and 52.

$$\begin{aligned} 51. \int \sec \theta d\theta & \quad 52. \int \csc \theta d\theta \end{aligned}$$

8.6

Integral Tables and Computer Algebra Systems

As we have studied, the basic techniques of integration are substitution and integration by parts. We apply these techniques to transform unfamiliar integrals into integrals whose forms we recognize or can find in a table. But where do the integrals in the tables come from? They come from applying substitutions and integration by parts, or by differentiating important functions that arise in practice or applications and tabling the results (as we did in creating Table 8.1). When an integral matches an integral in the table or can be changed into one of the tabulated integrals with some appropriate combination of algebra, trigonometry, substitution, and calculus, the matched result can be used to solve the integration problem at hand.

Computer Algebra Systems (CAS) can also be used to evaluate an integral, if such a system is available. However, beware that there are many relatively simple functions, like $\sin(x^2)$ or $1/\ln x$ for which even the most powerful computer algebra systems cannot find explicit antiderivative formulas because no such formulas exist.

In this section we discuss how to use tables and computer algebra systems to evaluate integrals.

Integral Tables

A Brief Table of Integrals is provided at the back of the book, after the index. (More extensive tables appear in compilations such as *CRC Mathematical Tables*, which contain thousands of integrals.) The integration formulas are stated in terms of constants a , b , c , m , n , and so on. These constants can usually assume any real value and need not be integers. Occasional limitations on their values are stated with the formulas. Formula 5 requires $n \neq -1$, for example, and Formula 11 requires $n \neq 2$.

The formulas also assume that the constants do not take on values that require dividing by zero or taking even roots of negative numbers. For example, Formula 8 assumes that $a \neq 0$, and Formulas 13(a) and (b) cannot be used unless b is positive.

The integrals in Examples 1–5 of this section can be evaluated using algebraic manipulation, substitution, or integration by parts. Here we illustrate how the integrals are found using the Brief Table of Integrals.

EXAMPLE 1 Find

$$\int x(2x + 5)^{-1} dx.$$

Solution We use Formula 8 (not 7, which requires $n \neq -1$):

$$\int x(ax + b)^{-1} dx = \frac{x}{a} - \frac{b}{a^2} \ln |ax + b| + C.$$

With $a = 2$ and $b = 5$, we have

$$\int x(2x + 5)^{-1} dx = \frac{x}{2} - \frac{5}{4} \ln |2x + 5| + C. \quad \blacksquare$$

EXAMPLE 2 Find

$$\int \frac{dx}{x\sqrt{2x+4}}.$$

Solution We use Formula 13(b):

$$\int \frac{dx}{x\sqrt{ax+b}} = \frac{1}{\sqrt{b}} \ln \left| \frac{\sqrt{ax+b} - \sqrt{b}}{\sqrt{ax+b} + \sqrt{b}} \right| + C, \quad \text{if } b > 0.$$

With $a = 2$ and $b = 4$, we have

$$\begin{aligned} \int \frac{dx}{x\sqrt{2x+4}} &= \frac{1}{\sqrt{4}} \ln \left| \frac{\sqrt{2x+4} - \sqrt{4}}{\sqrt{2x+4} + \sqrt{4}} \right| + C \\ &= \frac{1}{2} \ln \left| \frac{\sqrt{2x+4} - 2}{\sqrt{2x+4} + 2} \right| + C. \end{aligned}$$

Formula 13(a), which would require $b < 0$ here, is not appropriate in Example 2. It is appropriate, however, in the next example.

EXAMPLE 3 Find

$$\int \frac{dx}{x\sqrt{2x-4}}.$$

Solution We use Formula 13(a):

$$\int \frac{dx}{x\sqrt{ax-b}} = \frac{2}{\sqrt{b}} \tan^{-1} \sqrt{\frac{ax-b}{b}} + C.$$

With $a = 2$ and $b = 4$, we have

$$\int \frac{dx}{x\sqrt{2x-4}} = \frac{2}{\sqrt{4}} \tan^{-1} \sqrt{\frac{2x-4}{4}} + C = \tan^{-1} \sqrt{\frac{x-2}{2}} + C.$$

EXAMPLE 4 Find

$$\int \frac{dx}{x^2\sqrt{2x-4}}.$$

Solution We begin with Formula 15:

$$\int \frac{dx}{x^2\sqrt{ax+b}} = -\frac{\sqrt{ax+b}}{bx} - \frac{a}{2b} \int \frac{dx}{x\sqrt{ax+b}} + C.$$

With $a = 2$ and $b = -4$, we have

$$\int \frac{dx}{x^2\sqrt{2x-4}} = -\frac{\sqrt{2x-4}}{-4x} + \frac{2}{2 \cdot 4} \int \frac{dx}{x\sqrt{2x-4}} + C.$$

We then use Formula 13(a) to evaluate the integral on the right (Example 3) to obtain

$$\int \frac{dx}{x^2\sqrt{2x-4}} = \frac{\sqrt{2x-4}}{4x} + \frac{1}{4} \tan^{-1} \sqrt{\frac{x-2}{2}} + C.$$

EXAMPLE 5 Find

$$\int x \sin^{-1} x \, dx.$$

Solution We use Formula 99:

$$\int x^n \sin^{-1} ax \, dx = \frac{x^{n+1}}{n+1} \sin^{-1} ax - \frac{a}{n+1} \int \frac{x^{n+1} dx}{\sqrt{1-a^2x^2}}, \quad n \neq -1.$$

With $n = 1$ and $a = 1$, we have

$$\int x \sin^{-1} x \, dx = \frac{x^2}{2} \sin^{-1} x - \frac{1}{2} \int \frac{x^2 dx}{\sqrt{1-x^2}}.$$

The integral on the right is found in the table as Formula 33:

$$\int \frac{x^2}{\sqrt{a^2-x^2}} dx = \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) - \frac{1}{2} x \sqrt{a^2-x^2} + C.$$

With $a = 1$,

$$\int \frac{x^2}{\sqrt{1-x^2}} dx = \frac{1}{2} \sin^{-1} x - \frac{1}{2} x \sqrt{1-x^2} + C.$$

The combined result is

$$\begin{aligned} \int x \sin^{-1} x \, dx &= \frac{x^2}{2} \sin^{-1} x - \frac{1}{2} \left(\frac{1}{2} \sin^{-1} x - \frac{1}{2} x \sqrt{1-x^2} + C \right) \\ &= \left(\frac{x^2}{2} - \frac{1}{4} \right) \sin^{-1} x + \frac{1}{4} x \sqrt{1-x^2} + C'. \end{aligned}$$

Reduction Formulas

The time required for repeated integrations by parts can sometimes be shortened by applying formulas like

$$\int \tan^n x \, dx = \frac{1}{n-1} \tan^{n-1} x - \int \tan^{n-2} x \, dx \quad (1)$$

$$\int (\ln x)^n dx = x(\ln x)^n - n \int (\ln x)^{n-1} dx \quad (2)$$

$$\int \sin^n x \cos^m x \, dx = -\frac{\sin^{n-1} x \cos^{m+1} x}{m+n} + \frac{n-1}{m+n} \int \sin^{n-2} x \cos^m x \, dx \quad (n \neq -m). \quad (3)$$

Formulas like these are called **reduction formulas** because they replace an integral containing some power of a function with an integral of the same form with the power reduced. By applying such a formula repeatedly, we can eventually express the original integral in terms of a power low enough to be evaluated directly.

EXAMPLE 6 Using a Reduction Formula

Find

$$\int \tan^5 x \, dx.$$

Solution We apply Equation (1) with $n = 5$ to get

$$\int \tan^5 x \, dx = \frac{1}{4} \tan^4 x - \int \tan^3 x \, dx.$$

We then apply Equation (1) again, with $n = 3$, to evaluate the remaining integral:

$$\int \tan^3 x \, dx = \frac{1}{2} \tan^2 x - \int \tan x \, dx = \frac{1}{2} \tan^2 x + \ln |\cos x| + C.$$

The combined result is

$$\int \tan^5 x \, dx = \frac{1}{4} \tan^4 x - \frac{1}{2} \tan^2 x - \ln |\cos x| + C'. \quad \blacksquare$$

As their form suggests, reduction formulas are derived by integration by parts.

EXAMPLE 7 Deriving a Reduction Formula

Show that for any positive integer n ,

$$\int (\ln x)^n \, dx = x(\ln x)^n - n \int (\ln x)^{n-1} \, dx.$$

Solution We use the integration by parts formula

$$\int u \, dv = uv - \int v \, du$$

with

$$u = (\ln x)^n, \quad du = n(\ln x)^{n-1} \frac{dx}{x}, \quad dv = dx, \quad v = x,$$

to obtain

$$\int (\ln x)^n \, dx = x(\ln x)^n - n \int (\ln x)^{n-1} \, dx. \quad \blacksquare$$

Sometimes two reduction formulas come into play.

EXAMPLE 8 Find

$$\int \sin^2 x \cos^3 x \, dx.$$

Solution 1 We apply Equation (3) with $n = 2$ and $m = 3$ to get

$$\begin{aligned} \int \sin^2 x \cos^3 x \, dx &= -\frac{\sin x \cos^4 x}{2+3} + \frac{1}{2+3} \int \sin^0 x \cos^3 x \, dx \\ &= -\frac{\sin x \cos^4 x}{5} + \frac{1}{5} \int \cos^3 x \, dx. \end{aligned}$$

We can evaluate the remaining integral with Formula 61 (another reduction formula):

$$\int \cos^n ax \, dx = \frac{\cos^{n-1} ax \sin ax}{na} + \frac{n-1}{n} \int \cos^{n-2} ax \, dx.$$

With $n = 3$ and $a = 1$, we have

$$\begin{aligned}\int \cos^3 x \, dx &= \frac{\cos^2 x \sin x}{3} + \frac{2}{3} \int \cos x \, dx \\ &= \frac{\cos^2 x \sin x}{3} + \frac{2}{3} \sin x + C.\end{aligned}$$

The combined result is

$$\begin{aligned}\int \sin^2 x \cos^3 x \, dx &= -\frac{\sin x \cos^4 x}{5} + \frac{1}{5} \left(\frac{\cos^2 x \sin x}{3} + \frac{2}{3} \sin x + C \right) \\ &= -\frac{\sin x \cos^4 x}{5} + \frac{\cos^2 x \sin x}{15} + \frac{2}{15} \sin x + C' .\end{aligned}$$

Solution 2 Equation (3) corresponds to Formula 68 in the table, but there is another formula we might use, namely Formula 69. With $a = 1$, Formula 69 gives

$$\int \sin^n x \cos^m x \, dx = \frac{\sin^{n+1} x \cos^{m-1} x}{m+n} + \frac{m-1}{m+n} \int \sin^n x \cos^{m-2} x \, dx.$$

In our case, $n = 2$ and $m = 3$, so that

$$\begin{aligned}\int \sin^2 x \cos^3 x \, dx &= \frac{\sin^3 x \cos^2 x}{5} + \frac{2}{5} \int \sin^2 x \cos x \, dx \\ &= \frac{\sin^3 x \cos^2 x}{5} + \frac{2}{5} \left(\frac{\sin^3 x}{3} \right) + C \\ &= \frac{\sin^3 x \cos^2 x}{5} + \frac{2}{15} \sin^3 x + C.\end{aligned}$$

As you can see, it is faster to use Formula 69, but we often cannot tell beforehand how things will work out. Do not spend a lot of time looking for the “best” formula. Just find one that will work and forge ahead.

Notice also that Formulas 68 (Solution 1) and 69 (Solution 2) lead to different-looking answers. That is often the case with trigonometric integrals and is no cause for concern. The results are equivalent, and we may use whichever one we please. ■

Nonelementary Integrals

The development of computers and calculators that find antiderivatives by symbolic manipulation has led to a renewed interest in determining which antiderivatives can be expressed as finite combinations of elementary functions (the functions we have been studying) and which cannot. Integrals of functions that do not have elementary antiderivatives are called **nonelementary** integrals. They require infinite series (Chapter 11) or numerical methods for their evaluation. Examples of the latter include the error function (which measures the probability of random errors)

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} \, dt$$

and integrals such as

$$\int \sin x^2 \, dx \quad \text{and} \quad \int \sqrt{1+x^4} \, dx$$

that arise in engineering and physics. These and a number of others, such as

$$\int \frac{e^x}{x} dx, \quad \int e^{(e^x)} dx, \quad \int \frac{1}{\ln x} dx, \quad \int \ln(\ln x) dx, \quad \int \frac{\sin x}{x} dx,$$

$$\int \sqrt{1 - k^2 \sin^2 x} dx, \quad 0 < k < 1,$$

look so easy they tempt us to try them just to see how they turn out. It can be proved, however, that there is no way to express these integrals as finite combinations of elementary functions. The same applies to integrals that can be changed into these by substitution. The integrands all have antiderivatives, as a consequence of the Fundamental Theorem of the Calculus, Part 1, because they are continuous. However, none of the antiderivatives is elementary.

None of the integrals you are asked to evaluate in the present chapter falls into this category, but you may encounter nonelementary integrals in your other work.

Integration with a CAS

A powerful capability of computer algebra systems is their ability to integrate symbolically. This is performed with the **integrate command** specified by the particular system (for example, **int** in Maple, **Integrate** in Mathematica).

EXAMPLE 9 Using a CAS with a Named Function

Suppose that you want to evaluate the indefinite integral of the function

$$f(x) = x^2 \sqrt{a^2 + x^2}.$$

Using Maple, you first define or name the function:

$$> f := x^2 * \text{sqrt}(a^2 + x^2);$$

Then you use the integrate command on f , identifying the variable of integration:

$$> \text{int}(f, x);$$

Maple returns the answer

$$\frac{1}{4} x(a^2 + x^2)^{3/2} - \frac{1}{8} a^2 x \sqrt{a^2 + x^2} - \frac{1}{8} a^4 \ln(x + \sqrt{a^2 + x^2}).$$

If you want to see if the answer can be simplified, enter

$$> \text{simplify}(\%);$$

Maple returns

$$\frac{1}{8} a^2 x \sqrt{a^2 + x^2} + \frac{1}{4} x^3 \sqrt{a^2 + x^2} - \frac{1}{8} a^4 \ln(x + \sqrt{a^2 + x^2}).$$

If you want the definite integral for $0 \leq x \leq \pi/2$, you can use the format

$$> \text{int}(f, x = 0..Pi/2);$$

Maple (Version 5.1) will return the expression

$$\frac{1}{64} \pi (4a^2 + \pi^2)^{(3/2)} - \frac{1}{32} a^2 \pi \sqrt{4a^2 + \pi^2} + \frac{1}{8} a^4 \ln(2)$$

$$- \frac{1}{8} a^4 \ln(\pi + \sqrt{4a^2 + \pi^2}) + \frac{1}{16} a^4 \ln(a^2).$$

You can also find the definite integral for a particular value of the constant a :

```
> a:= 1;
> int(f, x = 0..1);
```

Maple returns the numerical answer

$$\frac{3}{8}\sqrt{2} + \frac{1}{8}\ln(\sqrt{2} - 1).$$

EXAMPLE 10 Using a CAS Without Naming the Function

Use a CAS to find

$$\int \sin^2 x \cos^3 x \, dx.$$

Solution With Maple, we have the entry

```
> int((sin^2)(x) * (cos^3)(x), x);
```

with the immediate return

$$-\frac{1}{5}\sin(x)\cos(x)^4 + \frac{1}{15}\cos(x)^2\sin(x) + \frac{2}{15}\sin(x).$$

EXAMPLE 11 A CAS May Not Return a Closed Form Solution

Use a CAS to find

$$\int (\cos^{-1} ax)^2 \, dx.$$

Solution Using Maple, we enter

```
> int((arccos(a * x))^2, x);
```

and Maple returns the expression

$$\int \arccos(ax)^2 \, dx,$$

indicating that it does not have a closed form solution known by Maple. In Chapter 11, you will see how series expansion may help to evaluate such an integral.

Computer algebra systems vary in how they process integrations. We used Maple 5.1 in Examples 9–11. Mathematica 4 would have returned somewhat different results:

1. In Example 9, given

```
In [1]:= Integrate [x^2 * Sqrt [a^2 + x^2], x]
```

Mathematica returns

$$\text{Out [1]} = \sqrt{a^2 + x^2} \left(\frac{a^2 x}{8} + \frac{x^3}{4} \right) - \frac{1}{8} a^4 \text{Log} [x + \sqrt{a^2 + x^2}]$$

without having to simplify an intermediate result. The answer is close to Formula 22 in the integral tables.